

COMSATS UNIVERSITY ISLAMABAD

WAH CAMPUS

Department of Computer Science

DevOps Lab Task #01

Course Title: DevOps

Course Code: CSC418

Instructor Name: Sir Ikram

Class: BSE-8A

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LINUX Commands:

Basic Linux directory navigation and management commands were implemented, including checking the current location, listing files, creating directories, and navigating between directories. The `projects/code` directory structure was also successfully created.

```
samikhan@osboxes:~$ pwd
/home/samikhan
samikhan@osboxes:~$ ls -lah
total 28K
drwxr-x--- 4 samikhan samikhan 4.0K Sep  7 05:22 .
drwxr-xr-x 4 root     root     4.0K Sep  7 05:17 ..
-rw-r--r-- 1 samikhan samikhan  220 Sep  7 05:17 .bash_logout
-rw-r--r-- 1 samikhan samikhan 3.7K Sep  7 05:17 .bashrc
drwx------ 3 samikhan samikhan 4.0K Sep  7 05:22 .local
-rw-r--r-- 1 samikhan samikhan  807 Sep  7 05:17 .profile
drwx------ 4 samikhan samikhan 4.0K Sep  7 05:22 snap
samikhan@osboxes:~$ ls
snap
samikhan@osboxes:~$ mkdir directory one
samikhan@osboxes:~$ ls
directory  one  snap
samikhan@osboxes:~$ cd /home/user/docs
-bash: cd: /home/user/docs: No such file or directory
samikhan@osboxes:~$ mkdir  -pv projects/code
mkdir: created directory 'projects'
mkdir: created directory 'projects/code'
```

File and directory management was implemented using commands such as `rmdir`, `touch`, `mkdir`, and `rm -rf`. The `useronefolder` directory was created, entered using `cd`, and a new `one.txt` file was created inside it.

```
samikhan@osboxes:~$ rmdir directory
samikhan@osboxes:~$ touch notes.txt
samikhan@osboxes:~$ mkdir folder
samikhan@osboxes:~$ rm -rf folder
samikhan@osboxes:~$  ls
notes.txt  one  projects  snap
samikhan@osboxes:~$ mkdir useronefolder
samikhan@osboxes:~$ mkdir usertwofolder
samikhan@osboxes:~$ ls
notes.txt  one  projects  snap  useronefolder  usertwofolder
samikhan@osboxes:~$ mkdir /useronefolder/fileone.txt
mkdir: cannot create directory ‘/useronefolder/fileone.txt’: No such file or dir
ectory
samikhan@osboxes:~$ pwd
/home/samikhan
samikhan@osboxes:~$ cd useronefolder
samikhan@osboxes:~/useronefolder$ one.txt
one.txt: command not found
samikhan@osboxes:~/useronefolder$ touch one.txt
samikhan@osboxes:~/useronefolder$
```

The useronefolder contents were copied to usertwofolder using sudo cp -r, which failed because the user account was not listed in the sudoers file. A new file.txt was then created with touch, and the ps aux command was used to list all currently running processes.

```
samikhan@osboxes:~/useronefolder$ touch one.txt
samikhan@osboxes:~/useronefolder$ ls
one.txt
samikhan@osboxes:~/useronefolder$ pwd
/home/samikhan/useronefolder
samikhan@osboxes:~/useronefolder$ sudo cp -r /home/samikhan/useronefolder/* /hom
e/samikhan/usertwofolder/
[sudo] password for samikhan:
samikhan is not in the sudoers file.
samikhan@osboxes:~/useronefolder$ touch file.txt
samikhan@osboxes:~/useronefolder$ ls
file.txt  one.txt
samikhan@osboxes:~/useronefolder$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.2  0.4  25028 15980 ?        Ss   05:06   0:05 /usr/lib/syst
root           2  0.0  0.0      0     0 ?        S    05:06   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    05:06   0:00 [pool_workque
root           4  0.0  0.0      0     0 ?        I<   05:06   0:00 [kworker/R-rc
root           5  0.0  0.0      0     0 ?        I<   05:06   0:00 [kworker/R-sy
root           6  0.0  0.0      0     0 ?        I<   05:06   0:00 [kworker/R-kv
root           7  0.0  0.0      0     0 ?        I<   05:06   0:00 [kworker/R-sl
root           8  0.0  0.0      0     0 ?        I<   05:06   0:00 [kworker/R-ne
root          11  0.0  0.0      0     0 ?        I<   05:06   0:00 [kworker/0:0H
root          12  0.0  0.0      0     0 ?        I    05:06   0:00 [kworker/u512
```

The output of ps aux continued to display system and kernel worker processes along with their resource usage. Network connectivity was then verified by pinging google.com four times with ping -c 4, confirming a successful response with 0% packet loss.

```
root       21494  0.0  0.0      0     0 ?        I    05:26   0:00 [kworker/1:0]
root       21499  0.0  0.0      0     0 ?        I    05:28   0:00 [kworker/u514
root       21528  0.0  0.0      0     0 ?        I<   05:33   0:00 [kworker/u516
root       21543  0.0  0.0      0     0 ?        I<   05:33   0:00 [kworker/u517
root       21545  0.0  0.0      0     0 ?        I<   05:33   0:00 [kworker/u517
root       21551  0.0  0.0      0     0 ?        I<   05:34   0:00 [kworker/u516
root       21566  0.0  0.0      0     0 ?        I    05:35   0:00 [kworker/u513
root       21569  0.0  0.0      0     0 ?        I    05:39   0:00 [kworker/u514
samikhan   21575  100  0.1  12536  4624 pts/1    R+   05:40   0:00 ps aux
samikhan@osboxes:~/useronefolder$ ping -c 4 google.com
PING google.com (142.250.202.142) 56(84) bytes of data.
64 bytes from pnfjra-al-in-f14.1e100.net (142.250.202.142): icmp_seq=1 ttl=128 t
ime=40.7 ms
64 bytes from pnfjra-al-in-f14.1e100.net (142.250.202.142): icmp_seq=2 ttl=128 t
ime=39.3 ms
64 bytes from pnfjra-al-in-f14.1e100.net (142.250.202.142): icmp_seq=3 ttl=128 t
ime=38.9 ms
64 bytes from pnfjra-al-in-f14.1e100.net (142.250.202.142): icmp_seq=4 ttl=128 t
ime=73.5 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 38.855/48.102/73.482/14.669 ms
samikhan@osboxes:~/useronefolder$
```

Network configuration details were inspected using the ip addr show command, which listed

the loopback and Ethernet interfaces along with their assigned addresses. The output confirmed the machine's IPv4 address, MAC address, and link status for the active network interface ens33.

```
ime=73.5 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 38.855/48.102/73.482/14.669 ms
samikhan@osboxes:~/useronefolder$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOW
N group default qlen 1000
    link/ether 00:0c:29:d7:f8:ff brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29d7f8ff
    inet 192.168.133.130/24 brd 192.168.133.255 scope global dynamic noprefixrou
te ens33
       valid_lft 1680sec preferred_lft 1680sec
    inet6 fe80::4eb2:6f91:af28:d74/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
samikhan@osboxes:~/useronefolder$
```

Disk space utilization was checked using the df -h command, which listed all mounted filesystems along with their total, used, and available space in human-readable format. The output confirmed sufficient free space on the root partition and other mounted volumes.

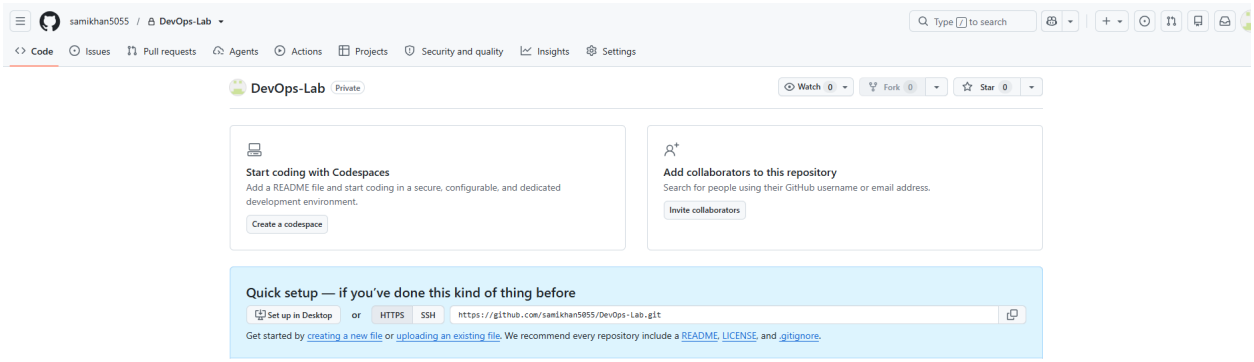
```
samikhan@osboxes:~/useronefolder$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           336M  2.4M  334M   1% /run
/dev/sda2       492G   13G  454G   3% /
tmpfs           1.7G     0  1.7G   0% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
tmpfs           1.7G   20K  1.7G   1% /tmp
tmpfs           336M  124K  336M   1% /run/user/1000
tmpfs           1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
tmpfs           336M   80K  336M   1% /run/user/1001
samikhan@osboxes:~/useronefolder$
```

A new file was opened in the nano text editor to begin writing a simple C++ program. The initial lines included the iostream header file and the std namespace declaration.

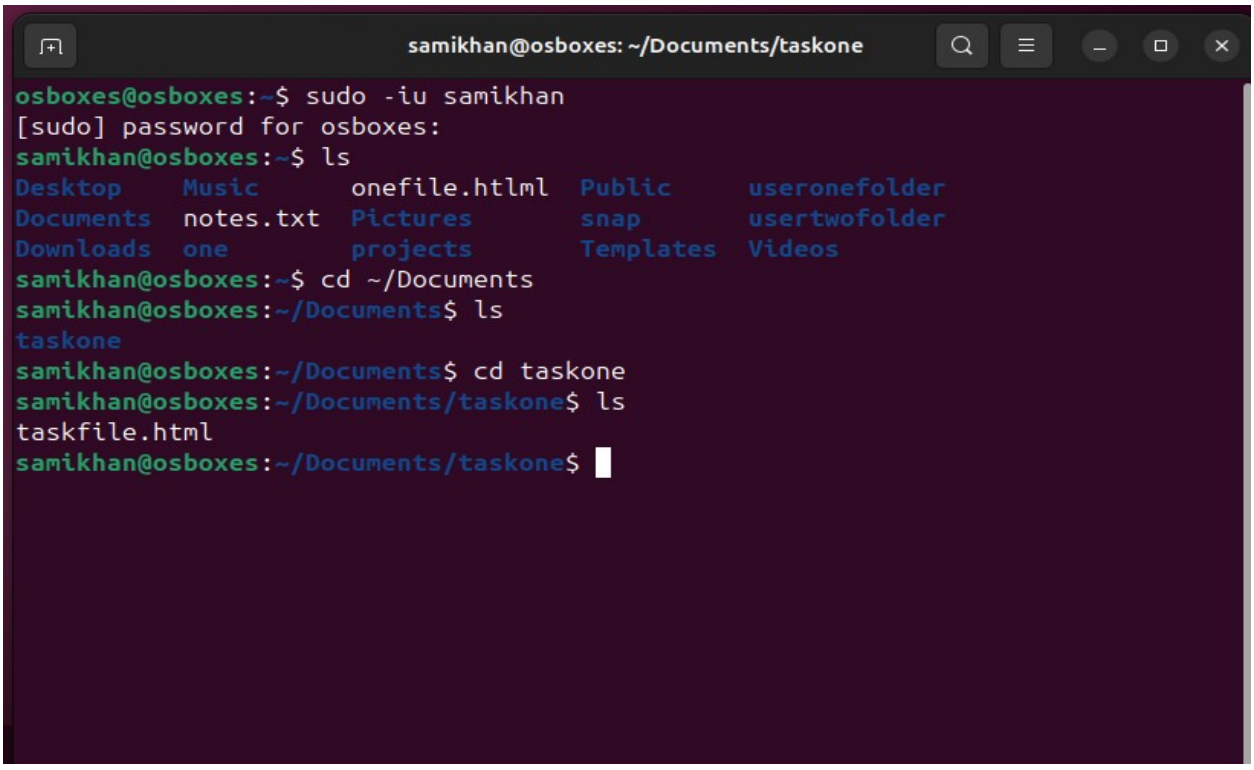
```
 GNU nano 8.3                    onefile.txt *
#include<iostream>
using namespace std;
```

Git Snapshots:

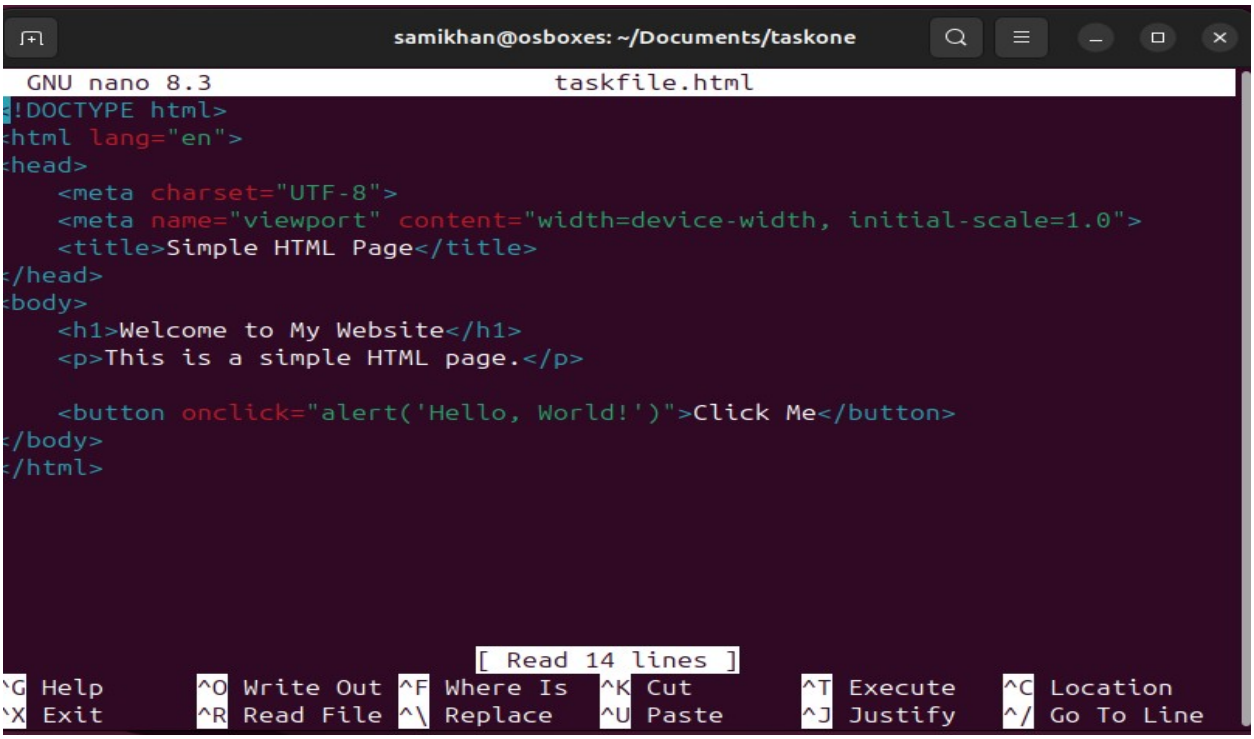
A new private GitHub repository named DevOps-Lab was created under the samikhan5055 account to host the project files. The repository's quick setup page displayed the HTTPS and SSH clone URLs along with options to start coding or add collaborators.



In the terminal, the samikhan user account was switched to using sudo -iu samikhan, and the home directory contents were listed. The Documents/taskone directory was located, containing a single file named taskfile.html.



The taskfile.html file was opened in the nano editor and edited to include a basic HTML structure with a title, a heading, a paragraph, and a button with an onclick alert event. This file was later used as the sample project file for the Git repository.



After returning to the taskone directory, the git init command was executed to initialize a new local Git repository. Git generated hints about the default branch name and suggested commands for renaming it if desired.

```
osboxes@osboxes:~$ sudo -iu samikhan
[sudo] password for osboxes:
samikhan@osboxes:~$ ls
Desktop    Music      onefile.htlml  Public     useronefolder
Documents  notes.txt  Pictures       snap       usertwofolder
Downloads  one        projects       Templates  Videos
samikhan@osboxes:~$ cd ~/Documents
samikhan@osboxes:~/Documents$ ls
taskone
samikhan@osboxes:~/Documents$ cd taskone
samikhan@osboxes:~/Documents/taskone$ ls
taskfile.html
samikhan@osboxes:~/Documents/taskone$ nano taskfile.html
samikhan@osboxes:~/Documents/taskone$ git init
hint: Using 'master' as the name for the initial branch. This default branch nam
e
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
```

Once the repository was initialized, git add . and git commit were attempted, but the commit failed because the Git user identity was not configured. The user.email and user.name values were then set using git config --global to resolve the identity error.

```
hint:   git branch -m <name>
Initialized empty Git repository in /home/samikhan/Documents/taskone/.git/
samikhan@osboxes:~/Documents/taskone$ git add .
samikhan@osboxes:~/Documents/taskone$ git commit -m "initial comment"
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'samikhan@osboxes.(none)')
samikhan@osboxes:~/Documents/taskone$ git config --global "samiiikhaan49@gmail.c
om"
error: invalid key: samiiikhaan49@gmail.com
samikhan@osboxes:~/Documents/taskone$ git config --global user.name "samikhan505
5"
git config --global user.email "samiiikhaan49@gmail.com"
samikhan@osboxes:~/Documents/taskone$ git config --global --list
```

With the Git identity configured, a git push was attempted but failed due to a missing remote destination, so the origin remote was added pointing to the GitHub repository. Subsequent push attempts to both main and master branches failed with a 'src refspec does not match' error since no commit had been made yet.

```
user.name=samikhan5055
user.email=samiiikhaan49@gmail.com
samikhan@osboxes:~/Documents/taskone$ git push
fatal: No configured push destination.
Either specify the URL from the command-line or configure a remote repository us
ing

    git remote add <name> <url>

and then push using the remote name

    git push <name>

samikhan@osboxes:~/Documents/taskone$ git remote add origin https://github.com/s
amikhan5055/DevOps-Lab.git
samikhan@osboxes:~/Documents/taskone$ git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/samikhan5055/DevOps-Lab.g
it'
samikhan@osboxes:~/Documents/taskone$ git branch --show-current
master
samikhan@osboxes:~/Documents/taskone$ git push -u origin master
error: src refspec master does not match any
error: failed to push some refs to 'https://github.com/samikhan5055/DevOps-Lab.g
```

The initial commit was successfully created, but pushing to GitHub failed first due to a missing upstream branch and then because GitHub no longer supports password authentication. To resolve this, an SSH key pair was generated using ssh-keygen -t ed25519 for secure authentication.

```
samikhan@osboxes:~/Documents/taskone$ git commit -m "initial commit"
[master (root-commit) 0015bf0] initial commit
 1 file changed, 14 insertions(+)
 create mode 100644 taskfile.html
samikhan@osboxes:~/Documents/taskone$ git push
fatal: The current branch master has no upstream branch.
To push the current branch and set the remote as upstream, use

    git push --set-upstream origin master

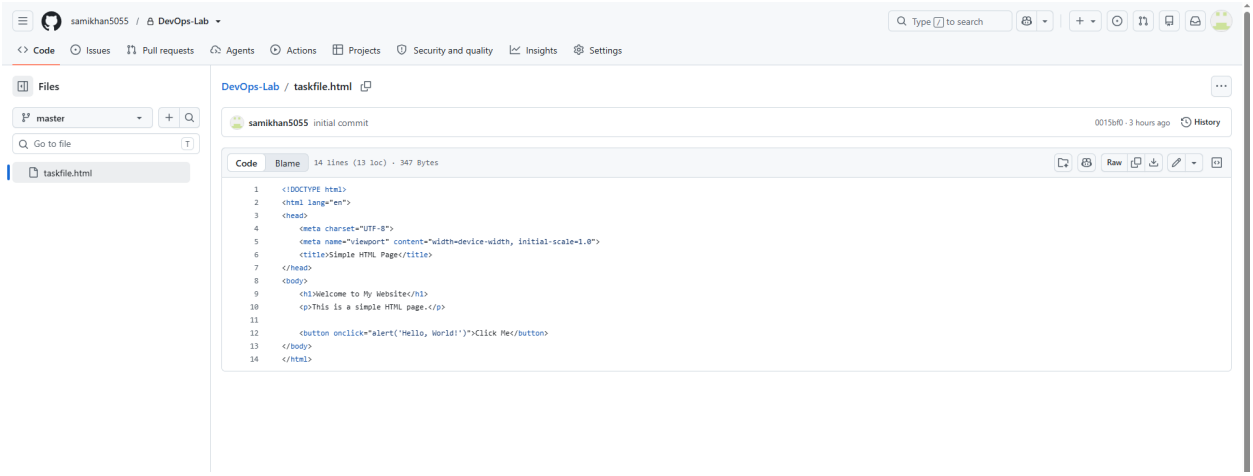
To have this happen automatically for branches without a tracking
upstream, see 'push.autoSetupRemote' in 'git help config'.

samikhan@osboxes:~/Documents/taskone$ git push --set-upstream origin master
Username for 'https://github.com': samikhan5055
Password for 'https://samikhan5055@github.com':
remote: Invalid username or token. Password authentication is not supported for
Git operations.
fatal: Authentication failed for 'https://github.com/samikhan5055/DevOps-Lab.git
/'
samikhan@osboxes:~/Documents/taskone$ ls ~/.ssh
samikhan@osboxes:~/Documents/taskone$ ssh-keygen -t ed25519 -C "samiiikhaan49@gm
ail.com"
```

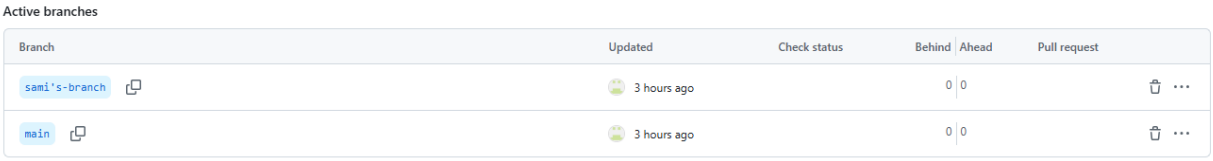
The public SSH key was displayed and added to the GitHub account, after which the remote URL was switched to use SSH instead of HTTPS. The repository was then successfully pushed to GitHub, the local branch was renamed to main, and pushed again to keep it in sync with the remote.

```
samikhan@osboxes:~/Documents/taskone$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIDv0gKTv1T7hbywFq86W1TL2HUI0dezBG8MnSRIdEMuZ
 samiiikhaan49@gmail.com
samikhan@osboxes:~/Documents/taskone$ git remote set-url origin git@github.com:s
amikhan5055/DevOps-Lab.git
samikhan@osboxes:~/Documents/taskone$ git push -u origin master
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 454 bytes | 454.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:samikhan5055/DevOps-Lab.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
samikhan@osboxes:~/Documents/taskone$ git branch -M main
samikhan@osboxes:~/Documents/taskone$ git push -u origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
```

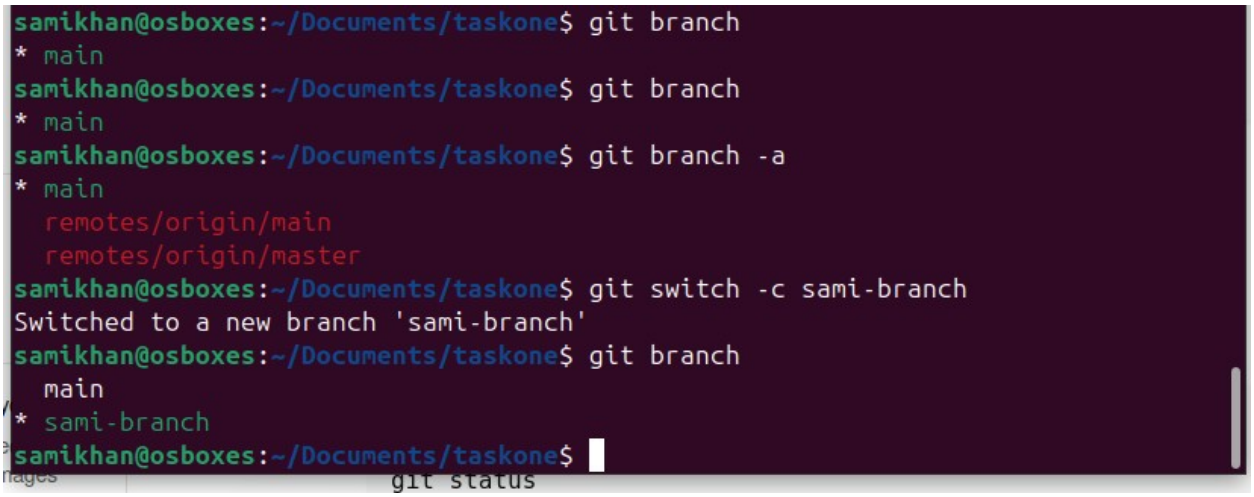
On GitHub, the DevOps-Lab repository confirmed that the taskfile.html file had been successfully pushed as the initial commit. The file's contents were visible directly in the repository's code view.



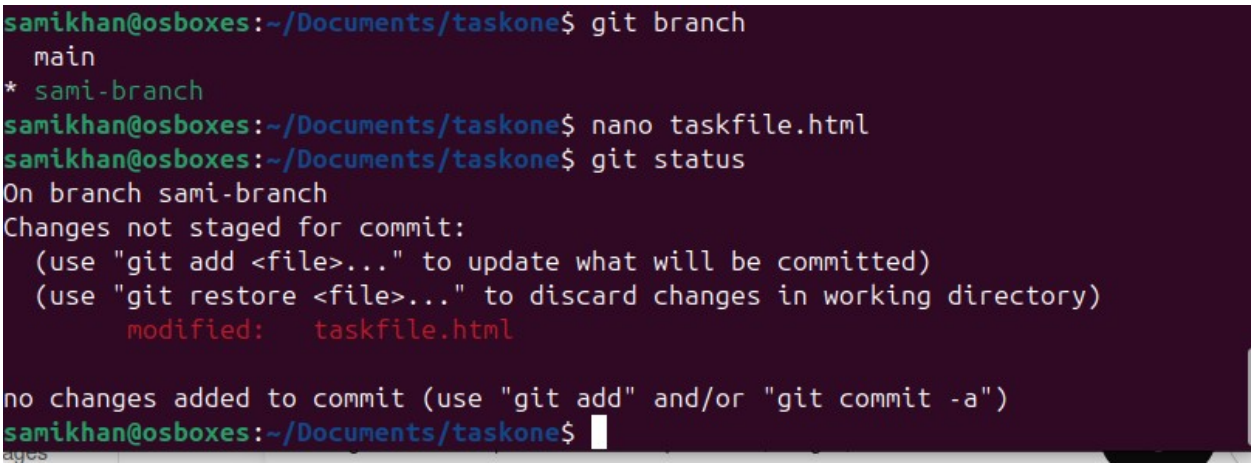
A new branch was created from the GitHub repository, resulting in two active branches: main and sami's-branch. The branches page showed both were up to date with zero commits ahead or behind each other.



Back in the terminal, git branch -a was used to confirm the presence of both local and remote branches. A new local branch named sami-branch was then created and switched to using git switch -c.



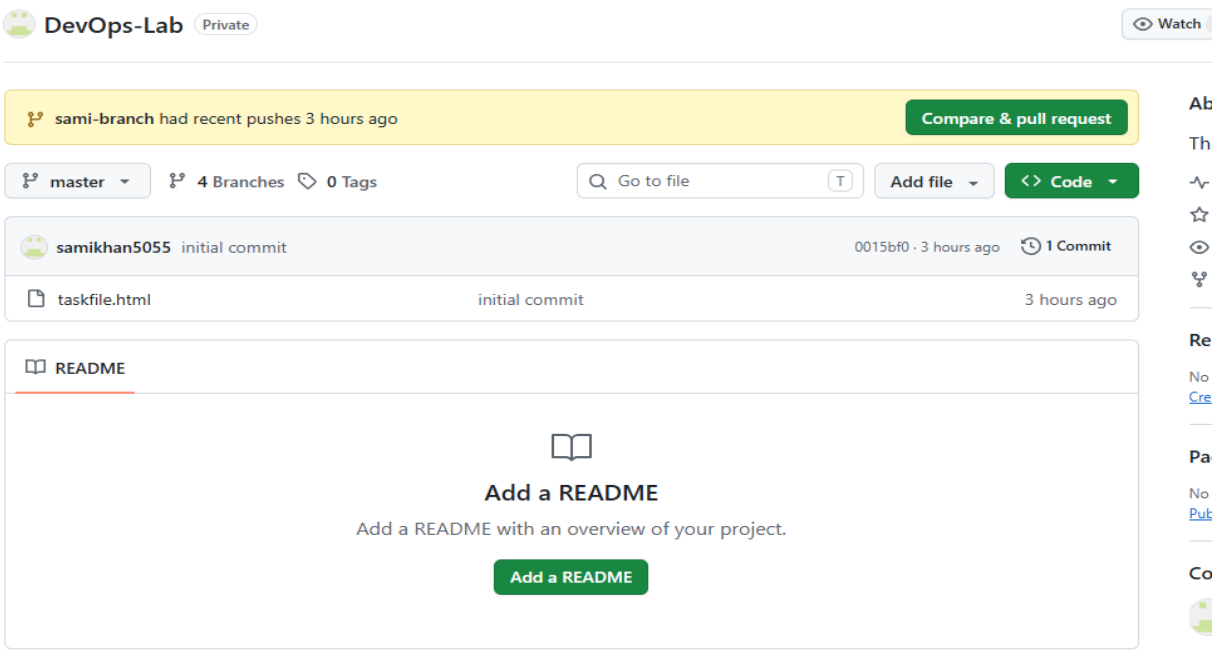
While on the sami-branch, the taskfile.html file was reopened in nano and modified. Running git status confirmed that the file had been changed but the modifications were not yet staged or committed.



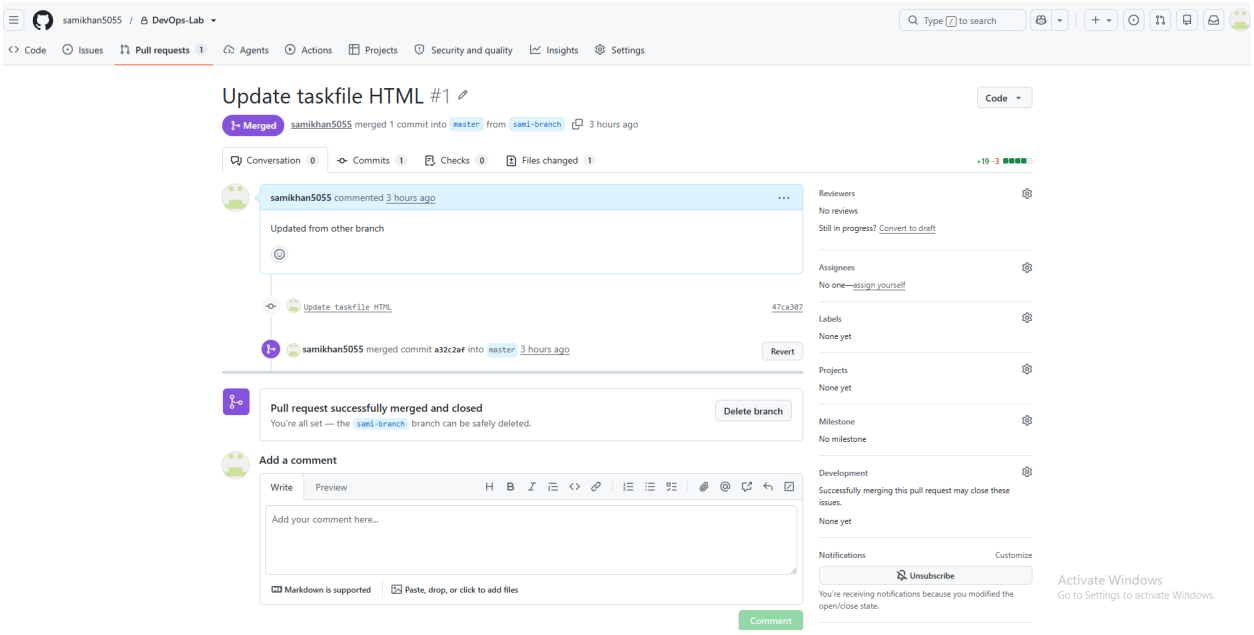
The modified file was staged and committed with the message "Update taskfile HTML", then pushed to the remote sami-branch using git push -u. GitHub responded with a link to create a pull request for merging the branch into main.

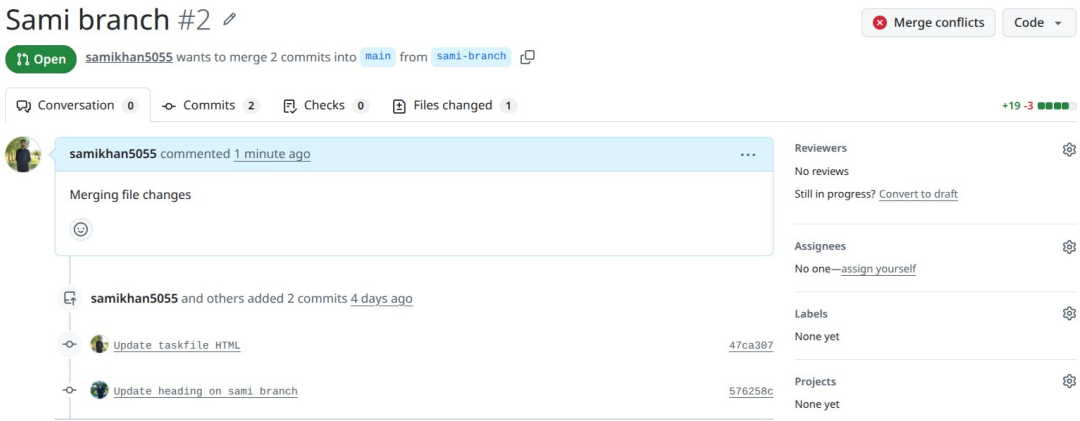
```
no changes added to commit (use "git add" and/or "git commit -a")
samikhan@osboxes:~/Documents/taskone$ git add taskfile.html
samikhan@osboxes:~/Documents/taskone$ git commit -m "Update taskfile HTML"
[sami-branch 47ca307] Update taskfile HTML
 1 file changed, 19 insertions(+), 3 deletions(-)
samikhan@osboxes:~/Documents/taskone$ git push -u origin sami-branch
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 629 bytes | 629.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'sami-branch' on GitHub by visiting:
remote:      https://github.com/samikhan5055/DevOps-Lab/pull/new/sami-branch
remote:
To github.com:samikhan5055/DevOps-Lab.git
 * [new branch]      sami-branch -> sami-branch
branch 'sami-branch' set up to track 'origin/sami-branch'.
samikhan@osboxes:~/Documents/taskone$
```

On GitHub, the repository detected the recent push to sami-branch and displayed a prompt to compare changes and open a pull request. This option was used to begin the process of merging the branch into master.

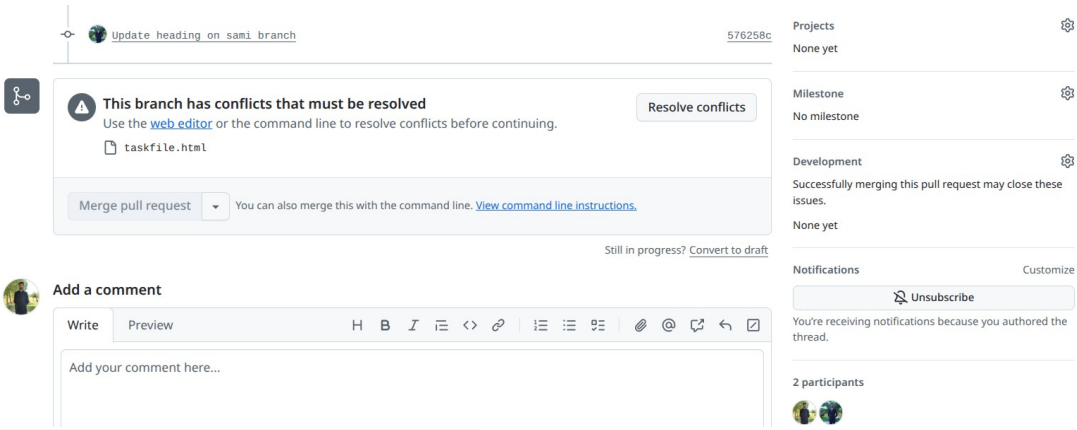


The pull request titled "Update taskfile HTML" was reviewed and merged into the master branch. GitHub confirmed the merge was completed successfully and offered the option to delete the now-merged sami-branch.





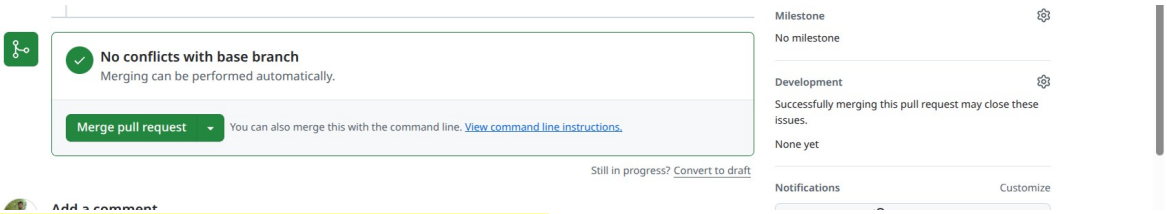
A merge conflict occurred when changes from the `sami-branch` were merged into `main` because both branches contained different changes to the same part of the code. The conflict was identified and needs to be resolved manually before completing the merge.



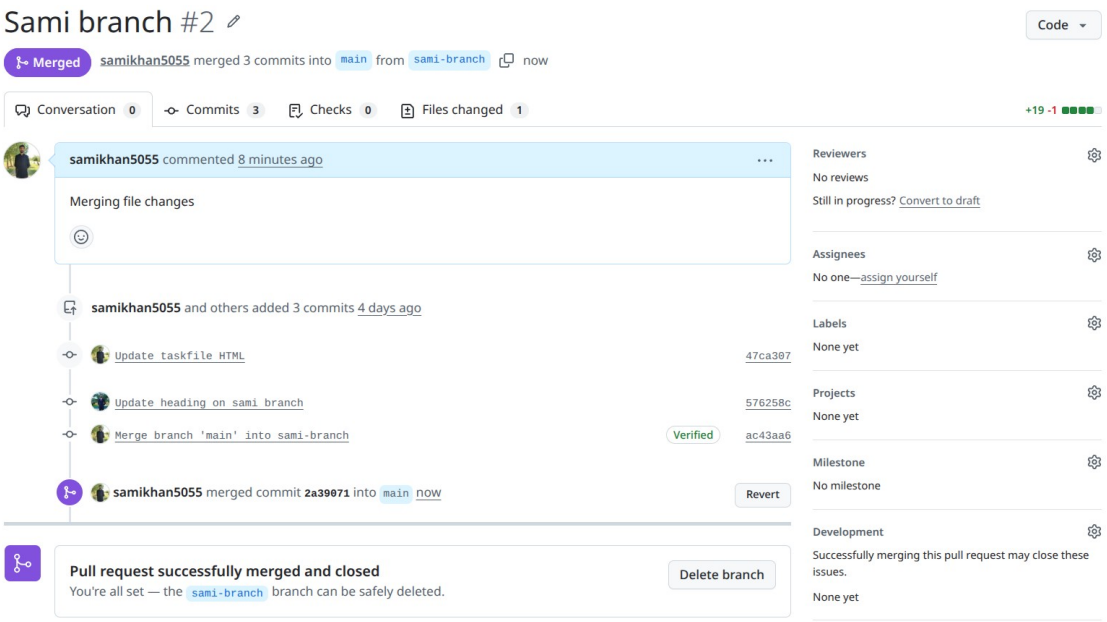
The merge conflict was successfully resolved by selecting and combining the required changes. The resolved code was committed and pushed, and the updated changes are now reflected in the repository.



Merging after resolving conflicts

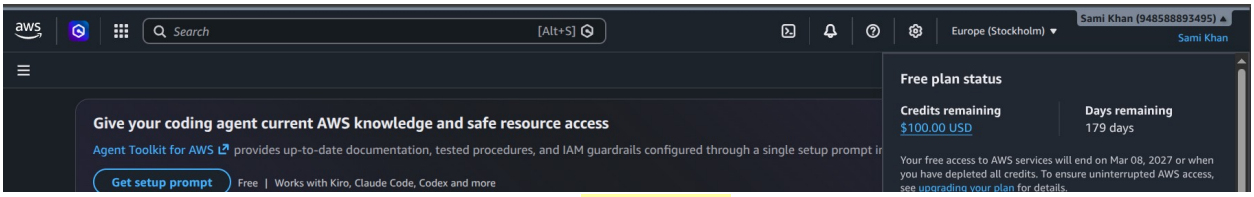


The pull request was successfully merged into the `main` branch, and the pull request was automatically closed after the merge was completed.

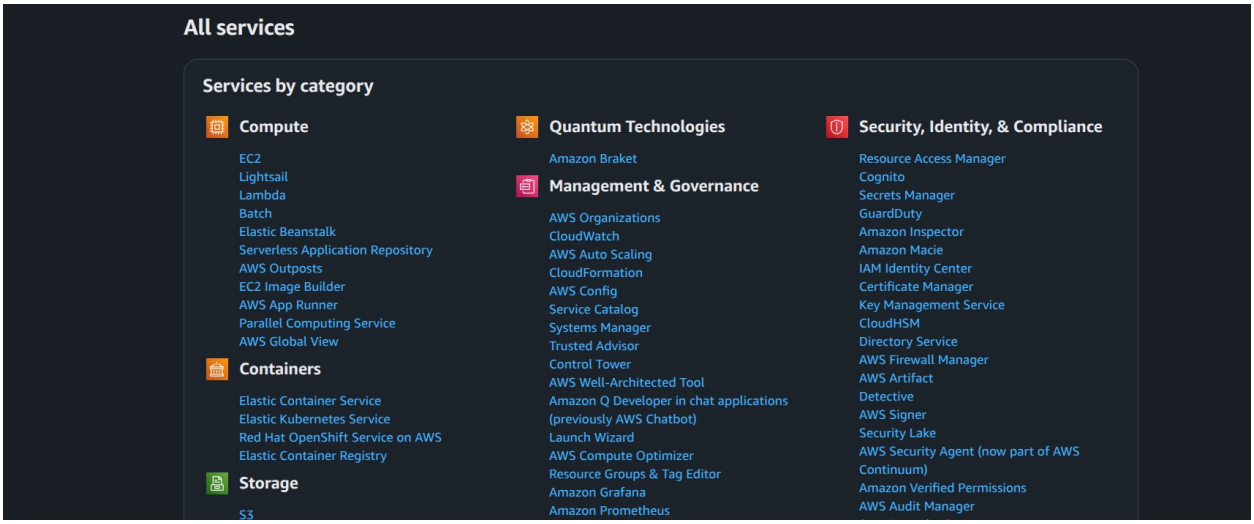


AWS Account Creation and Verification:

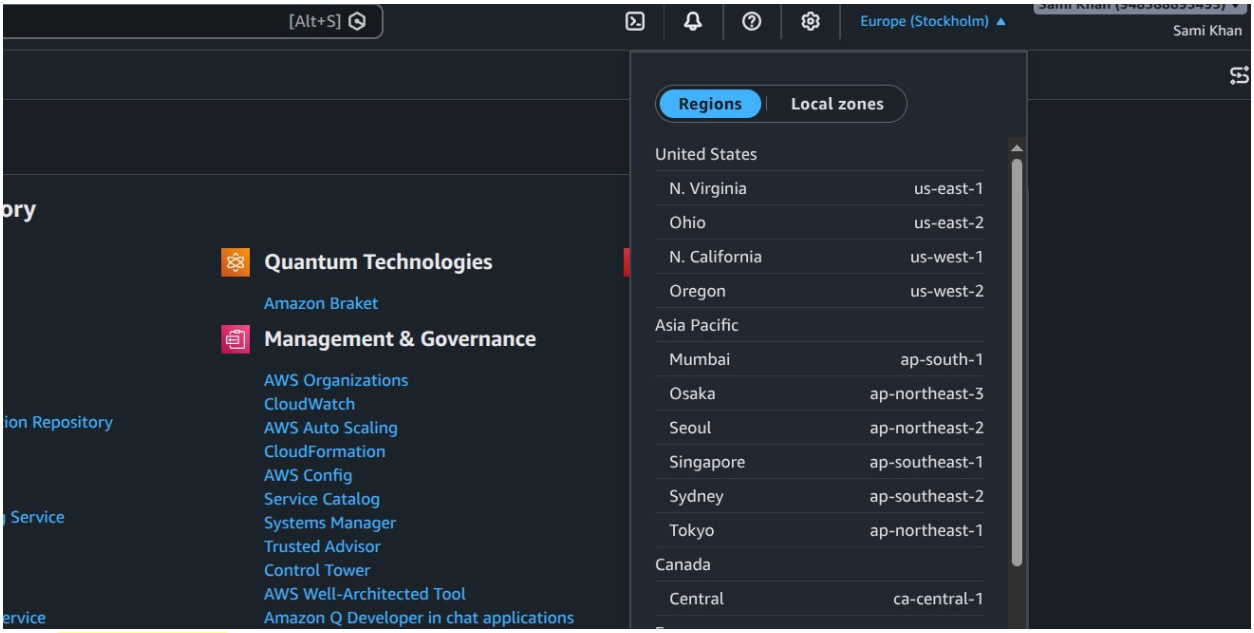
After signing in, the AWS Management Console confirmed successful account creation and verification under the free plan. The dashboard displayed the available credits and remaining days of free-tier access, verifying that the AWS account was fully active and ready for use.



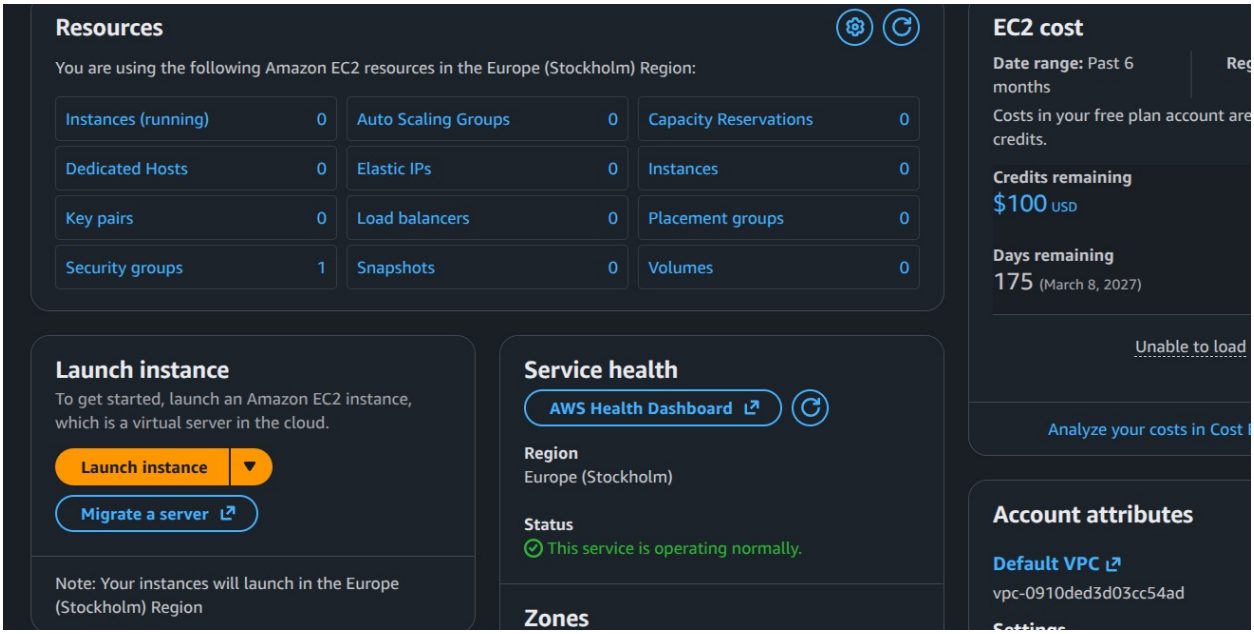
The AWS Management Console displays the **All Services** section, showing AWS services organized into categories such as Compute, Containers, Storage, Quantum Technologies, Management & Governance, and Security.



The AWS Region selection menu is displayed, showing available regions and their corresponding region codes across the United States, Asia Pacific, and Canada.

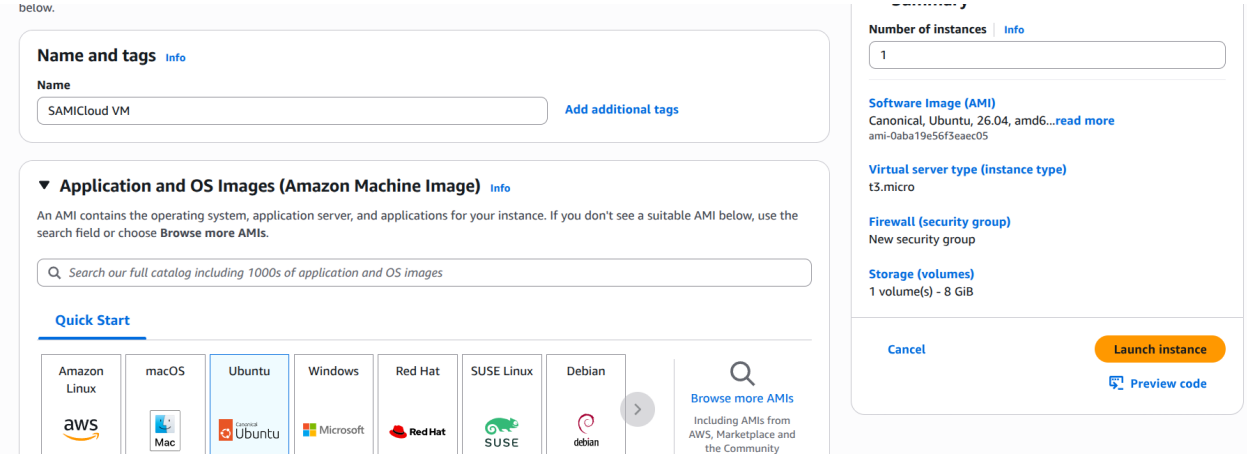


The Amazon EC2 dashboard shows the available resources in the Europe (Stockholm) Region, along with the option to launch a new EC2 instance. It also confirms that the AWS service is operating normally.

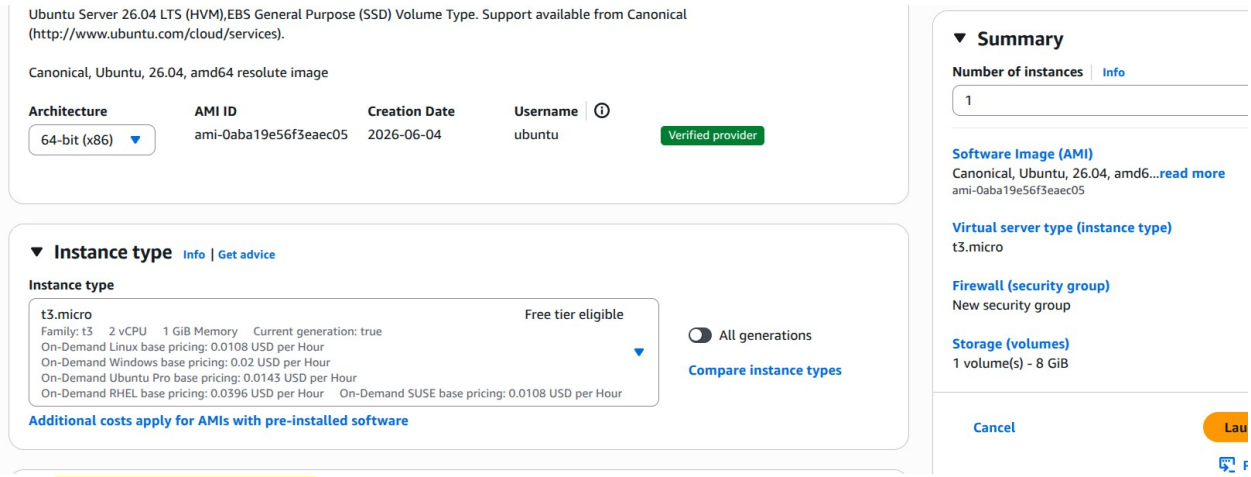


The EC2 instance configuration is completed with the name **SAMICloud VM** and Ubuntu 26.04 selected as the operating system image.
The instance type is set to **t3.micro**, with a new security group and **8 GiB** of storage configured.

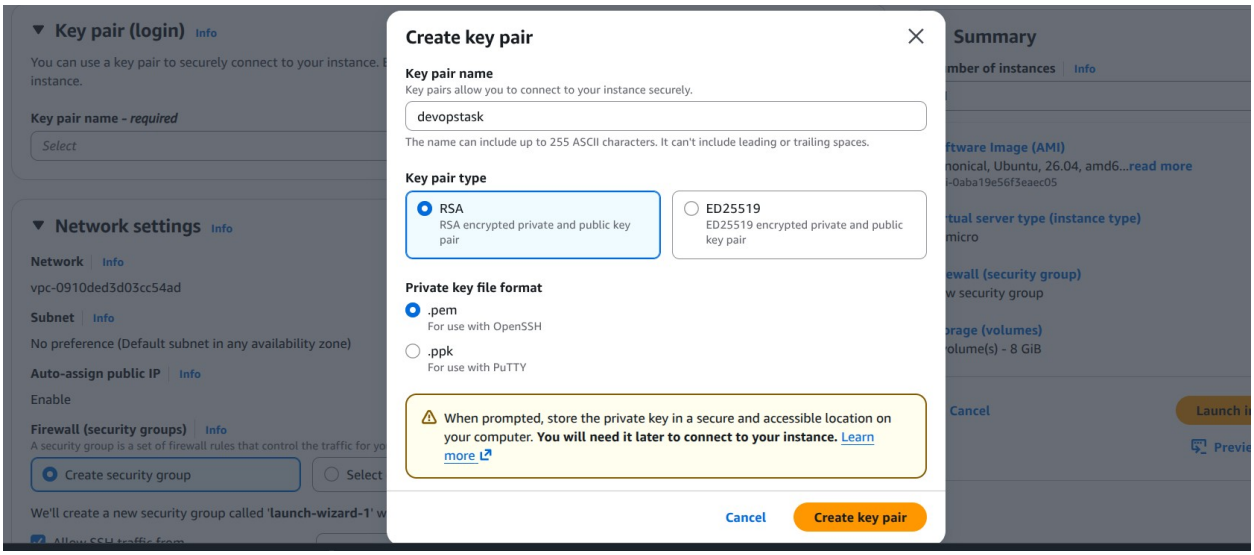
The summary confirms the selected settings and provides the option to launch the EC2 instance.



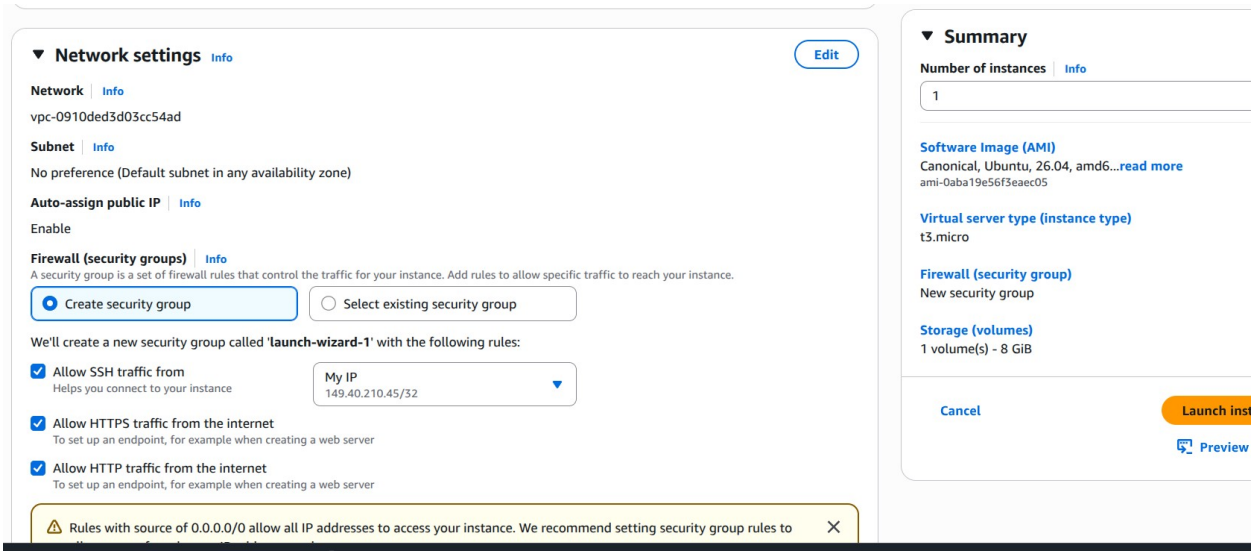
The Ubuntu Server 26.04 LTS AMI is selected with a **64-bit (x86)** architecture and the **t3.micro** instance type.
The t3.micro provides **2 vCPUs and 1 GiB memory** and is marked as **Free Tier eligible**.
The configuration summary confirms **1 instance**, a new security group, and **8 GiB** of storage.



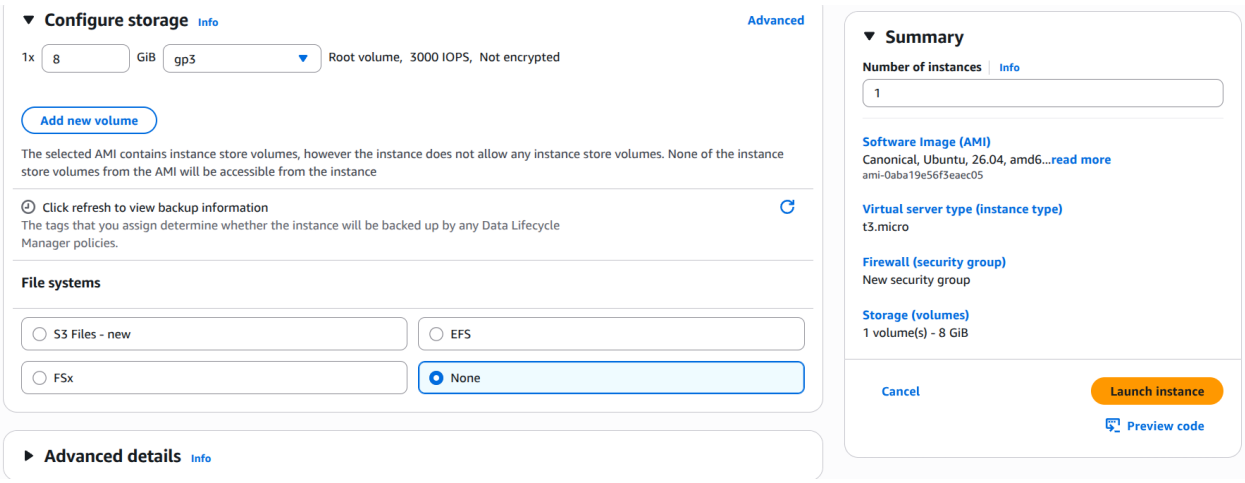
The **devopstask** key pair is being created using the **RSA** key type and **.pem** private key format for secure SSH access.
The private key will be securely stored and used later to connect to the EC2 instance.



The network settings are configured with a default VPC and **auto-assigned public IP** enabled for the EC2 instance.
A new security group is created with **SSH access from the user's IP**, along with HTTP and HTTPS traffic enabled from the internet.

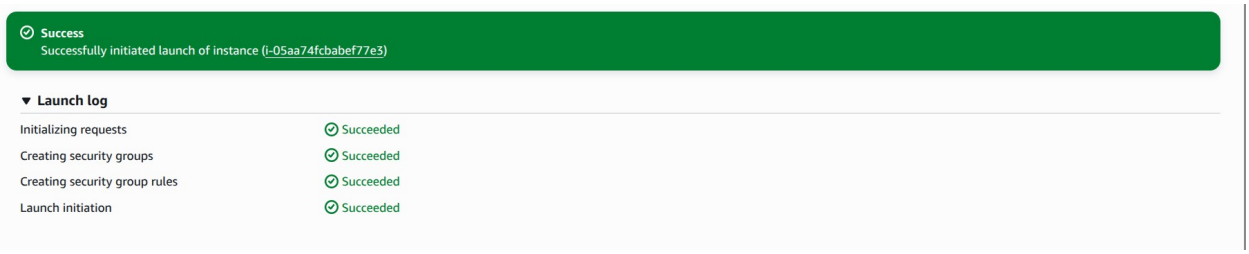


The storage configuration is set with an **8 GiB gp3 root volume** using 3,000 IOPS, while no additional file systems such as S3 Files, EFS, or FSx are configured. The instance is therefore configured with a single 8 GiB root storage volume and is ready for launch.

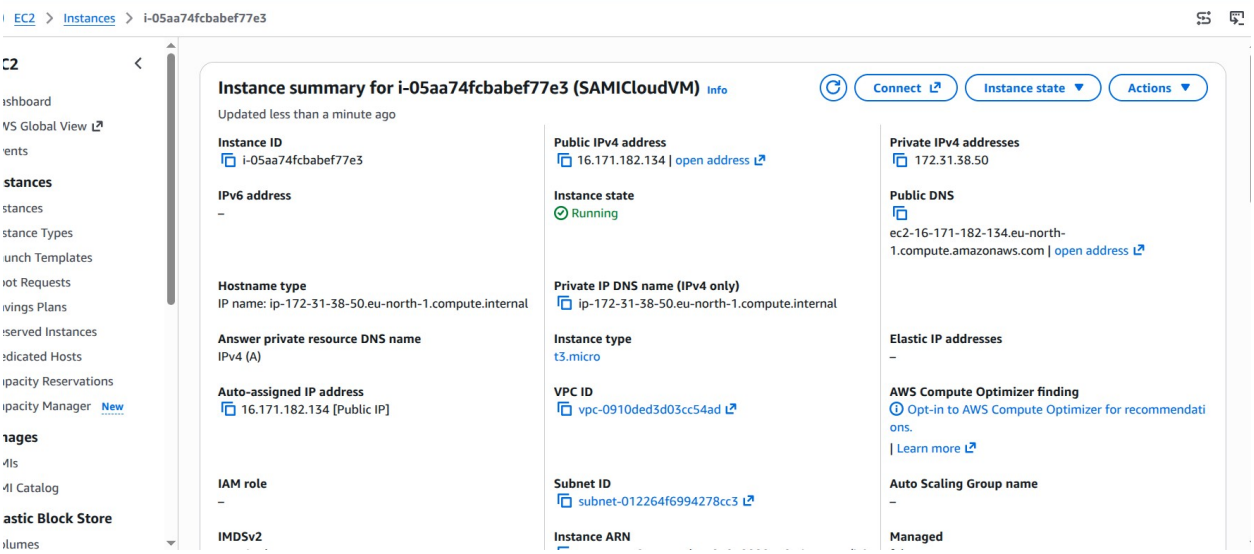


The EC2 instance was successfully launched, as confirmed by the green success message displaying the newly created instance ID. The launch process completed successfully, and the instance is now available for further configuration and access.

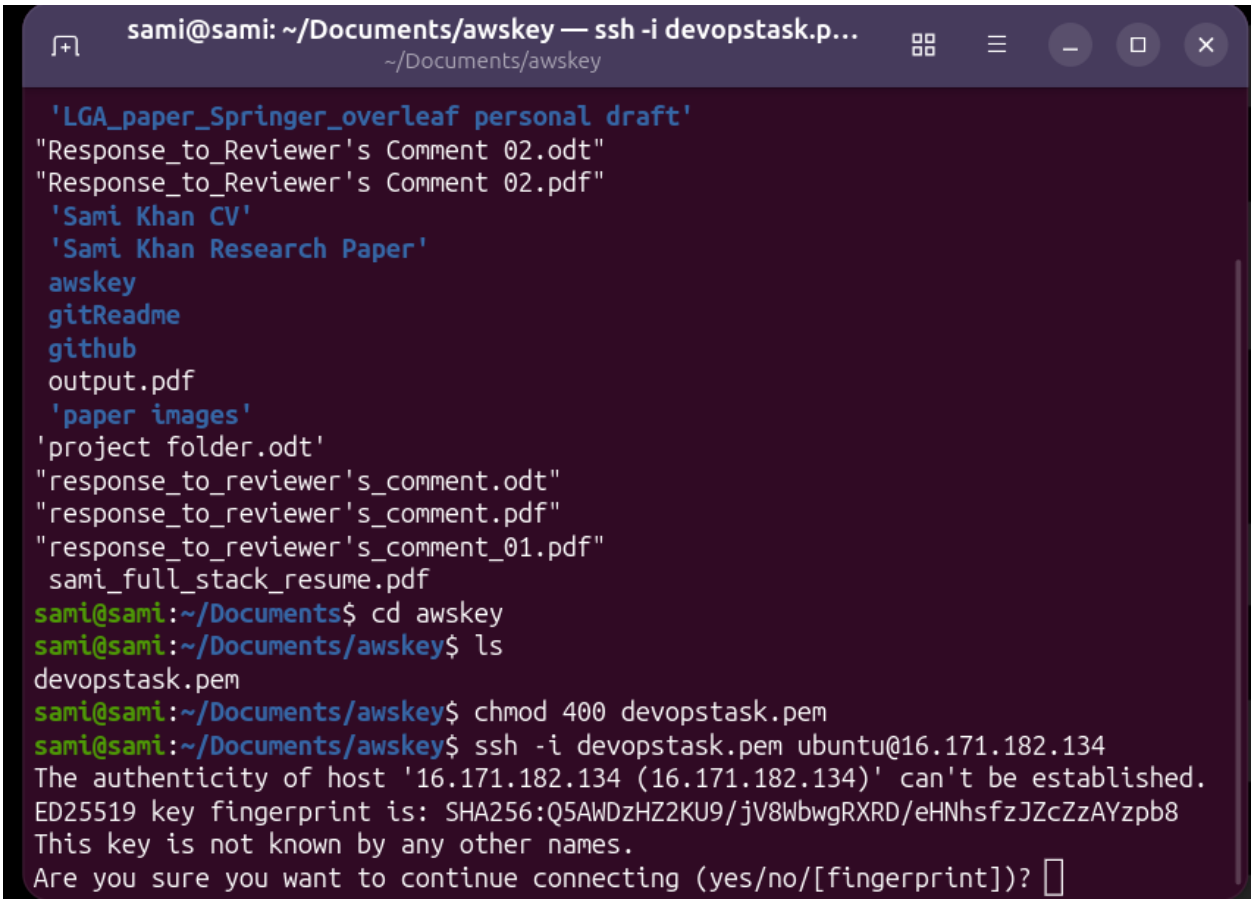




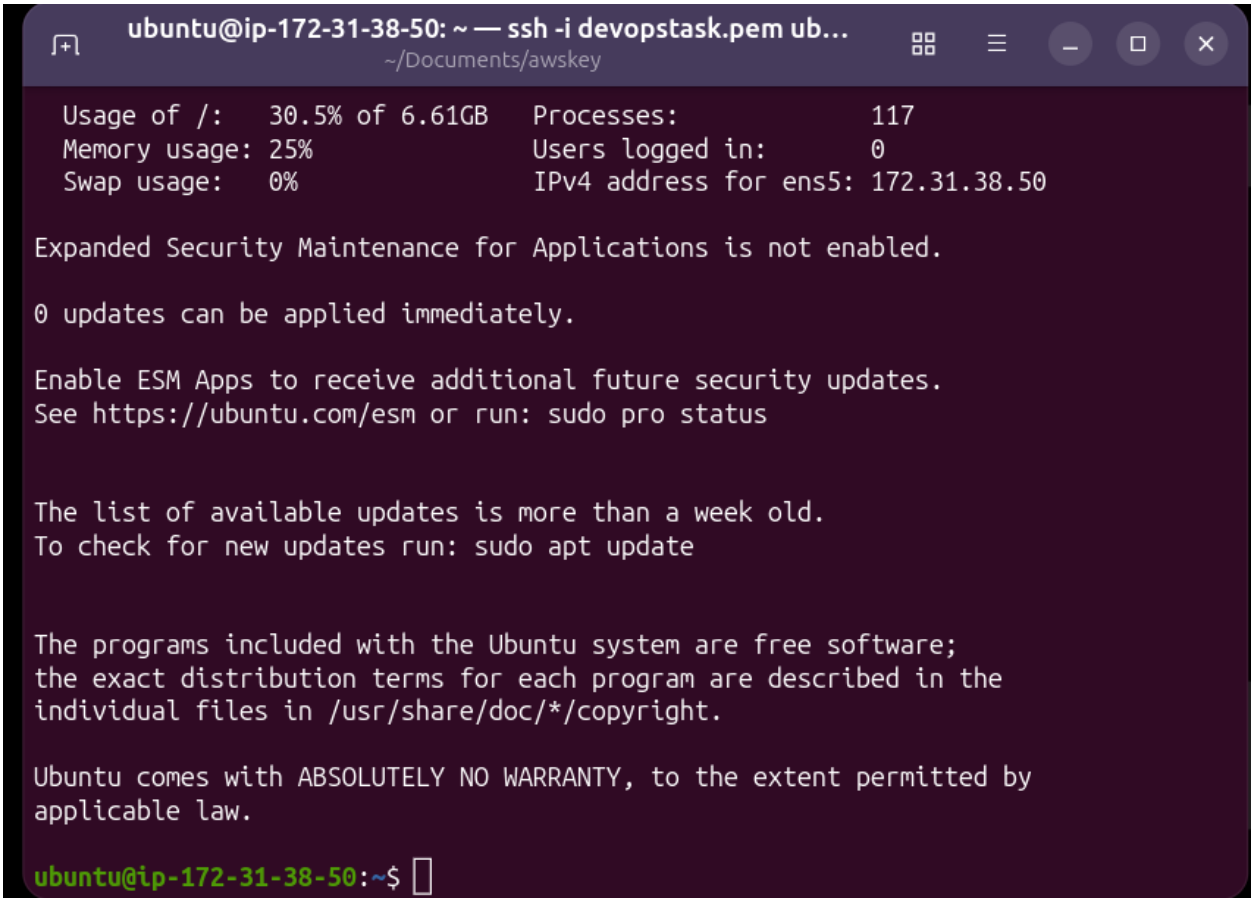
The EC2 instance **SAMICloudVM** is successfully running with instance ID `i-05aa74fcbabef77e3` and instance type **t3.micro**. It has a public IPv4 address **16.171.182.134** for external access and a private IP address **172.31.38.50** within the VPC. The instance is deployed in the default VPC and subnet, confirming that the virtual server is active and ready for connection.



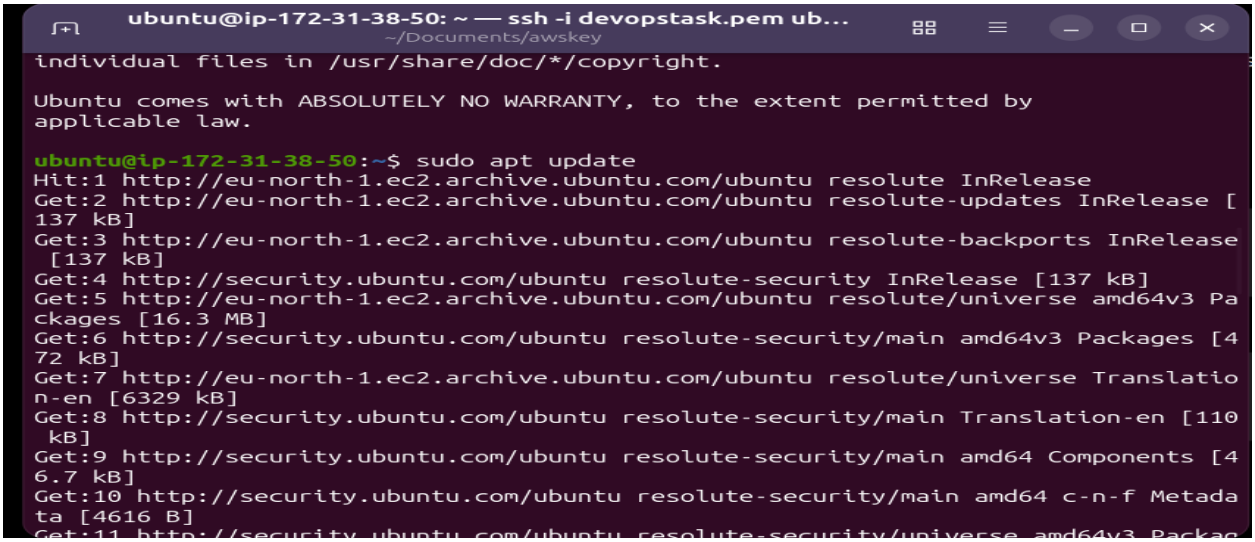
The terminal shows the `devopstask.pem` key being assigned secure permissions using `chmod 400`, followed by an SSH command connecting to the EC2 instance through its public IP address. The connection request was verified using the server's ED25519 fingerprint.



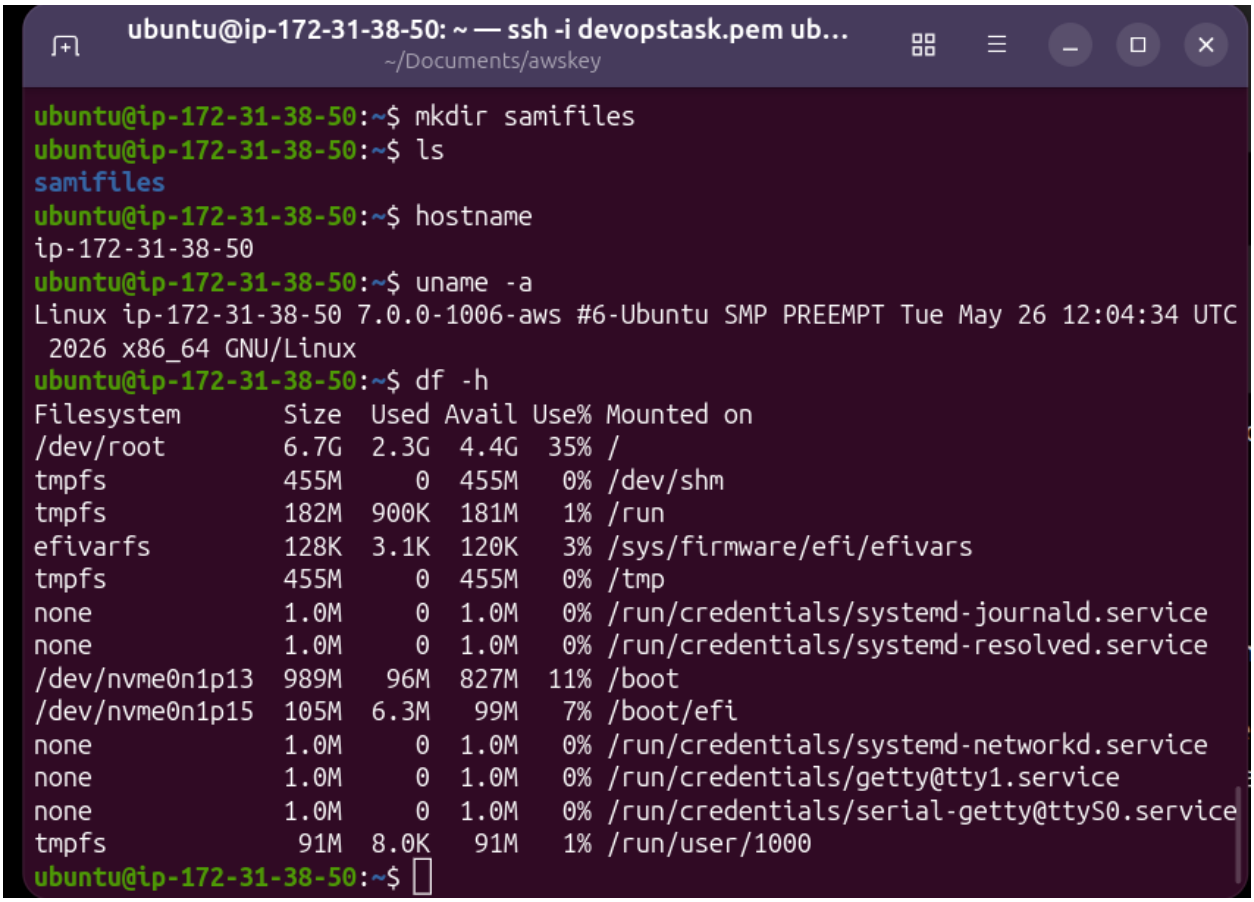
The terminal confirms a successful SSH connection to the **Ubuntu 26.04 LTS EC2 instance**. The system information displays the server's resource usage and private IP address, and the `ubuntu@ip-172-31-38-50:~$` prompt confirms that the remote server is ready for further commands.



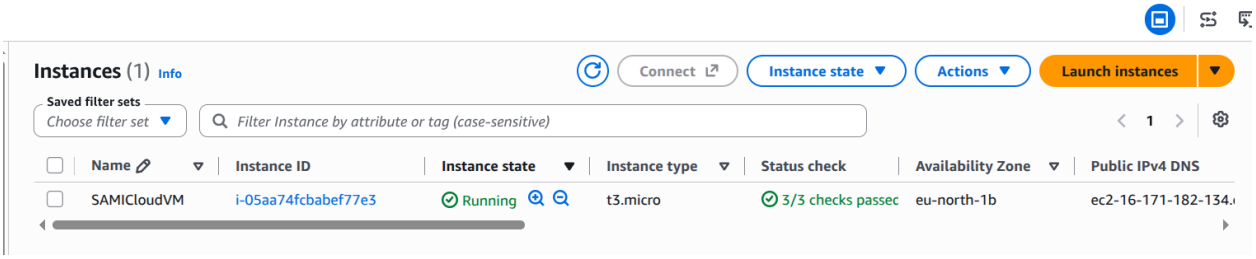
Run sdu apt update command on cloud base VM.

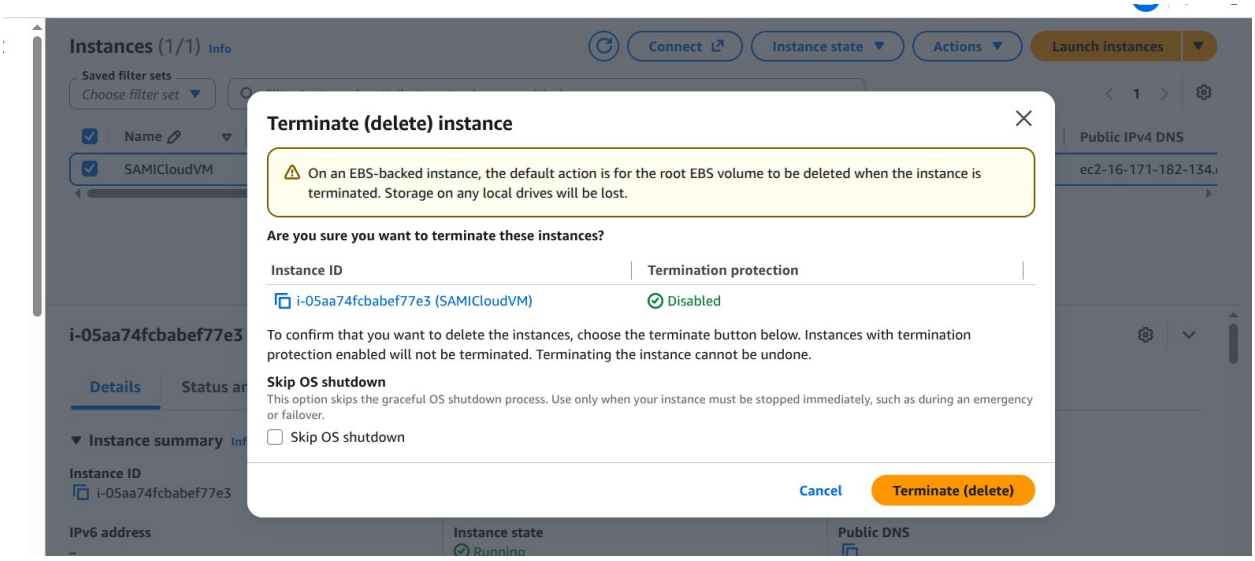


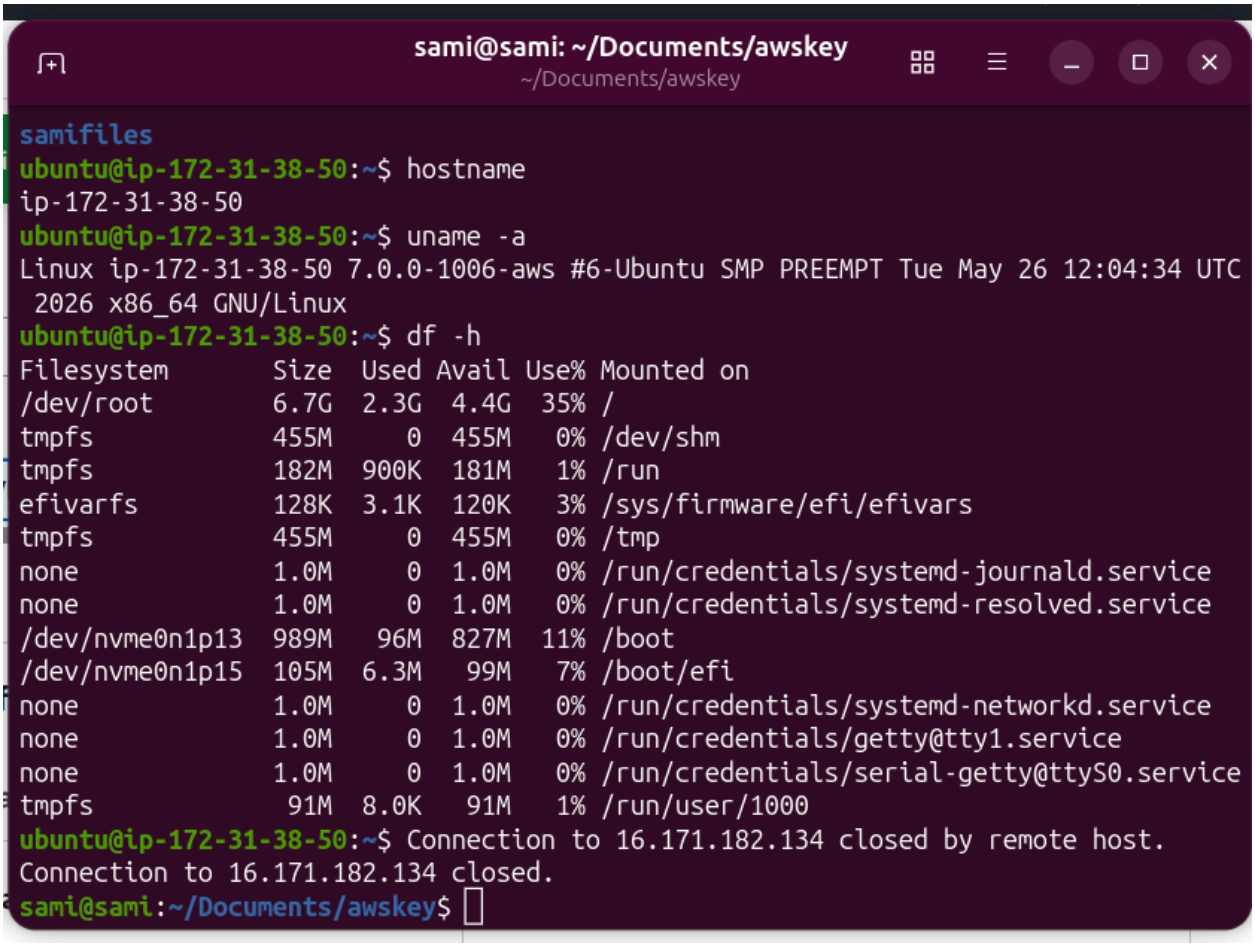
The screenshot shows the creation of a `samifiles` directory on the EC2 Ubuntu server using `mkdir`, followed by verification with `ls`. The `hostname`, `uname -a`, and `df -h` commands are also used to verify the server identity, Linux system information, and available disk storage.



Now in below screenshot we have EC2 instance details, its running.




Instances (1/1) Info
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Name
SAMICloudVM
Public IPv4 DNS
ec2-16-171-182-134.
Terminate (delete) instance
On an EBS-backed instance, the default action is for the root EBS volume to be deleted when the instance is terminated. Storage on any local drives will be lost.
Are you sure you want to terminate these instances?
Instance ID
Termination protection
i-05aa74fcbabef77e3 (SAMICloudVM)
Disabled
To confirm that you want to delete the instances, choose the terminate button below. Instances with termination protection enabled will not be terminated. Terminating the instance cannot be undone.
Skip OS shutdown
This option skips the graceful OS shutdown process. Use only when your instance must be stopped immediately, such as during an emergency or failover.
Skip OS shutdown
Cancel
Terminate (delete)
i-05aa74fcbabef77e3
Details
Instance summary
Instance ID
i-05aa74fcbabef77e3
IPv6 address
Instance state
Running
Public DNS


sami@sami: ~/Documents/awskey
~/Documents/awskey
samifiles
ubuntu@ip-172-31-38-50:~$ hostname
ip-172-31-38-50
ubuntu@ip-172-31-38-50:~$ uname -a
Linux ip-172-31-38-50 7.0.0-1006-aws #6-Ubuntu SMP PREEMPT Tue May 26 12:04:34 UTC 2026 x86_64 GNU/Linux
ubuntu@ip-172-31-38-50:~$ df -h
Filesystem Size Used Avail Use% Mounted on
/dev/root 6.7G 2.3G 4.4G 35% /
tmpfs 455M 0 455M 0% /dev/shm
tmpfs 182M 900K 181M 1% /run
efivarfs 128K 3.1K 120K 3% /sys/firmware/efi/efivars
tmpfs 455M 0 455M 0% /tmp
none 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service
none 1.0M 0 1.0M 0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13 989M 96M 827M 11% /boot
/dev/nvme0n1p15 105M 6.3M 99M 7% /boot/efi
none 1.0M 0 1.0M 0% /run/credentials/systemd-networkd.service
none 1.0M 0 1.0M 0% /run/credentials/getty@tty1.service
none 1.0M 0 1.0M 0% /run/credentials/serial-getty@ttyS0.service
tmpfs 91M 8.0K 91M 1% /run/user/1000
ubuntu@ip-172-31-38-50:~$ Connection to 16.171.182.134 closed by remote host.
Connection to 16.171.182.134 closed.
sami@sami:~/Documents/awskey$

Connect Cloud VM through ubuntu terminal also checked git version of that.

```
ubuntu@ip-172-31-42-28:~$ sudo apt install git
git is already the newest version (1:2.53.0-1ubuntu1).
git set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
ubuntu@ip-172-31-42-28:~$ git --version
git version 2.53.0
ubuntu@ip-172-31-42-28:~$ ls -la ~/.ssh
total 12
drwx------ 2 ubuntu ubuntu 4096 Sep 14 09:11 .
drwxr-x--- 4 ubuntu ubuntu 4096 Sep 14 09:20 ..
-rw------- 1 ubuntu ubuntu  392 Sep 14 09:11 authorized_keys
ubuntu@ip-172-31-42-28:~$ ssh-keygen -t ed25519 -C "samiiikhaan49@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519):
Enter passphrase for "/home/ubuntu/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519
```

Authenticated github account in cloud VM where ubuntu is installed

```
The key's randomart image is:
+--[ED25519 256]--+
|                 |
|                 |
|        o o      |
|     . + = o     |
|    = + S = . .  |
|     O = O . =   |
|    . =.O E o .  |
|     o.*o@ + .   |
|    ..*B*.*o+    |
+----[SHA256]-----+
ubuntu@ip-172-31-42-28:~$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIKW4b7KQ8wBrzxnGY2GRv4kH8gitFYk7mFSZnw1OnSvr
 samiiikhaan49@gmail.com
ubuntu@ip-172-31-42-28:~$ ssh -T git@github.com
The authenticity of host 'github.com (4.225.11.194)' can't be established.
ED25519 key fingerprint is: SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi samikhan5055! You've successfully authenticated, but GitHub does not provide
shell access.
ubuntu@ip-172-31-42-28:~$
```

Now repo is being cloned in cloud vm.

```
|    . =.O E o .  |
|     o.*o@ + .   |
|    ..*B*.*o+    |
+----[SHA256]-----+
ubuntu@ip-172-31-42-28:~$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIKW4b7KQ8wBrzxnGY2GRv4kH8gitFYk7mFSZnw1OnSvr
 samiiikhaan49@gmail.com
ubuntu@ip-172-31-42-28:~$ ssh -T git@github.com
The authenticity of host 'github.com (4.225.11.194)' can't be established.
ED25519 key fingerprint is: SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi samikhan5055! You've successfully authenticated, but GitHub does not provide
shell access.
ubuntu@ip-172-31-42-28:~$ git clone git@github.com:samikhan5055/DevOps-Lab.git
Cloning into 'DevOps-Lab'...
remote: Enumerating objects: 18, done.
remote: Counting objects: 100% (18/18), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 18 (delta 5), reused 9 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (18/18), 4.81 KiB | 2.40 MiB/s, done.
Resolving deltas: 100% (5/5), done.
ubuntu@ip-172-31-42-28:~$
```

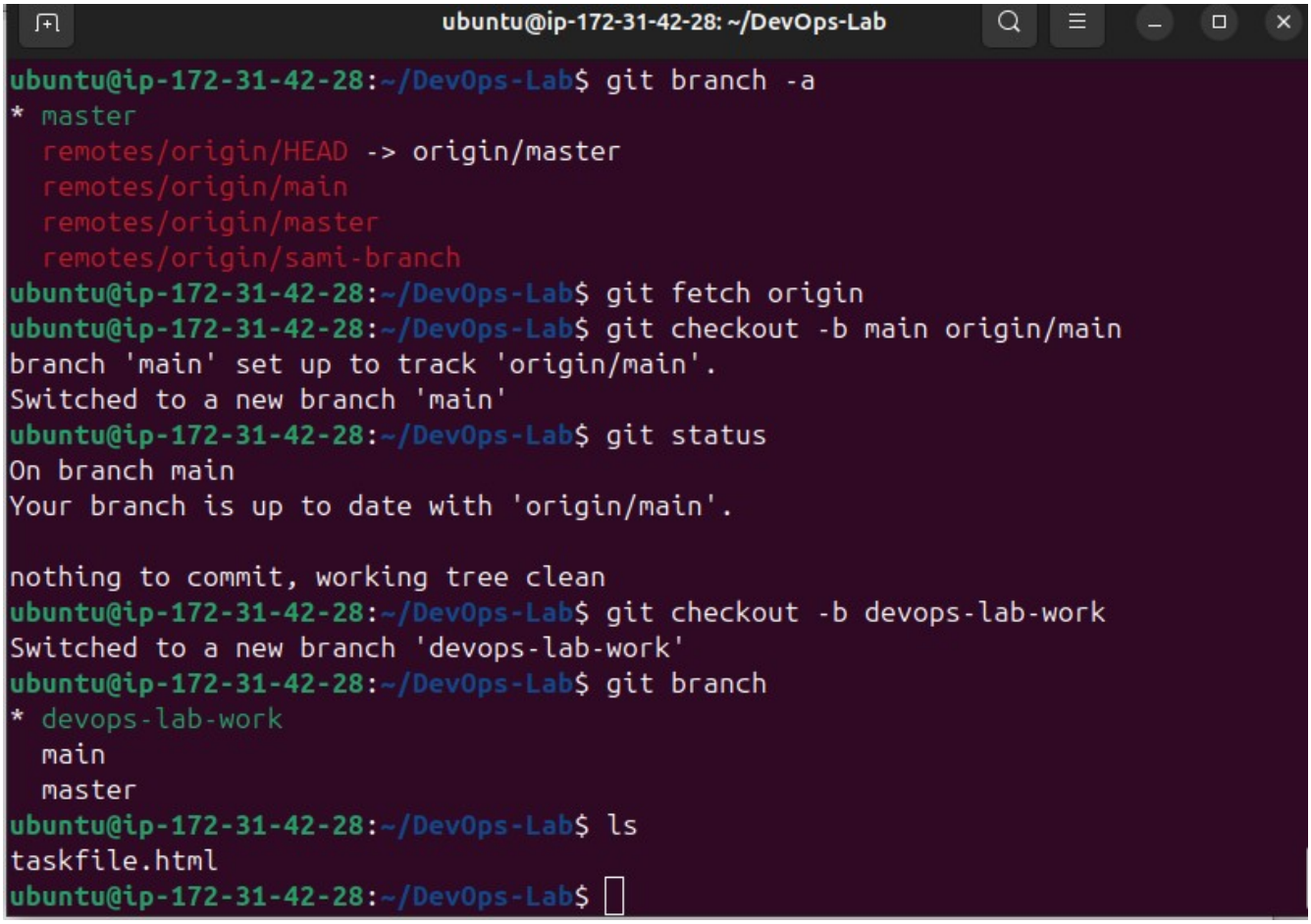
Now we have performed git operation on repo like status check, branches etc

```
ubuntu@ip-172-31-42-28: ~/DevOps-Lab
ubuntu@ip-172-31-42-28:~$ cd DevOps-Lab
ubuntu@ip-172-31-42-28:~/DevOps-Lab$ git status
On branch master
Your branch is up to date with 'origin/master'.

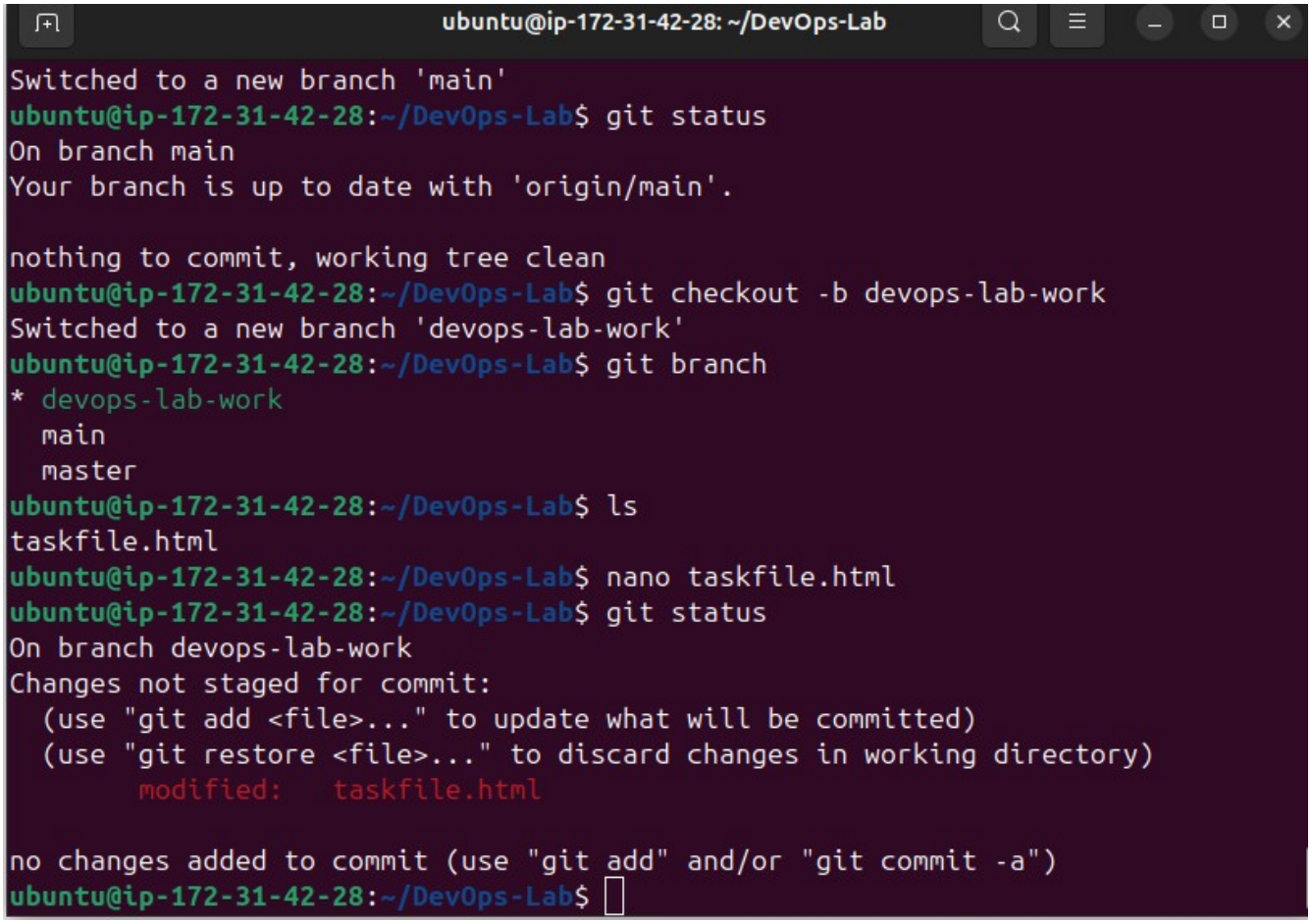
nothing to commit, working tree clean
ubuntu@ip-172-31-42-28:~/DevOps-Lab$ git remote -v
origin  git@github.com:samikhan5055/DevOps-Lab.git (fetch)
origin  git@github.com:samikhan5055/DevOps-Lab.git (push)
ubuntu@ip-172-31-42-28:~/DevOps-Lab$ git branch -a
* master
  remotes/origin/HEAD -> origin/master
  remotes/origin/main
  remotes/origin/master
  remotes/origin/sami-branch
ubuntu@ip-172-31-42-28:~/DevOps-Lab$ git fetch origin
ubuntu@ip-172-31-42-28:~/DevOps-Lab$ git checkout -b main origin/main
branch 'main' set up to track 'origin/main'.
Switched to a new branch 'main'
ubuntu@ip-172-31-42-28:~/DevOps-Lab$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

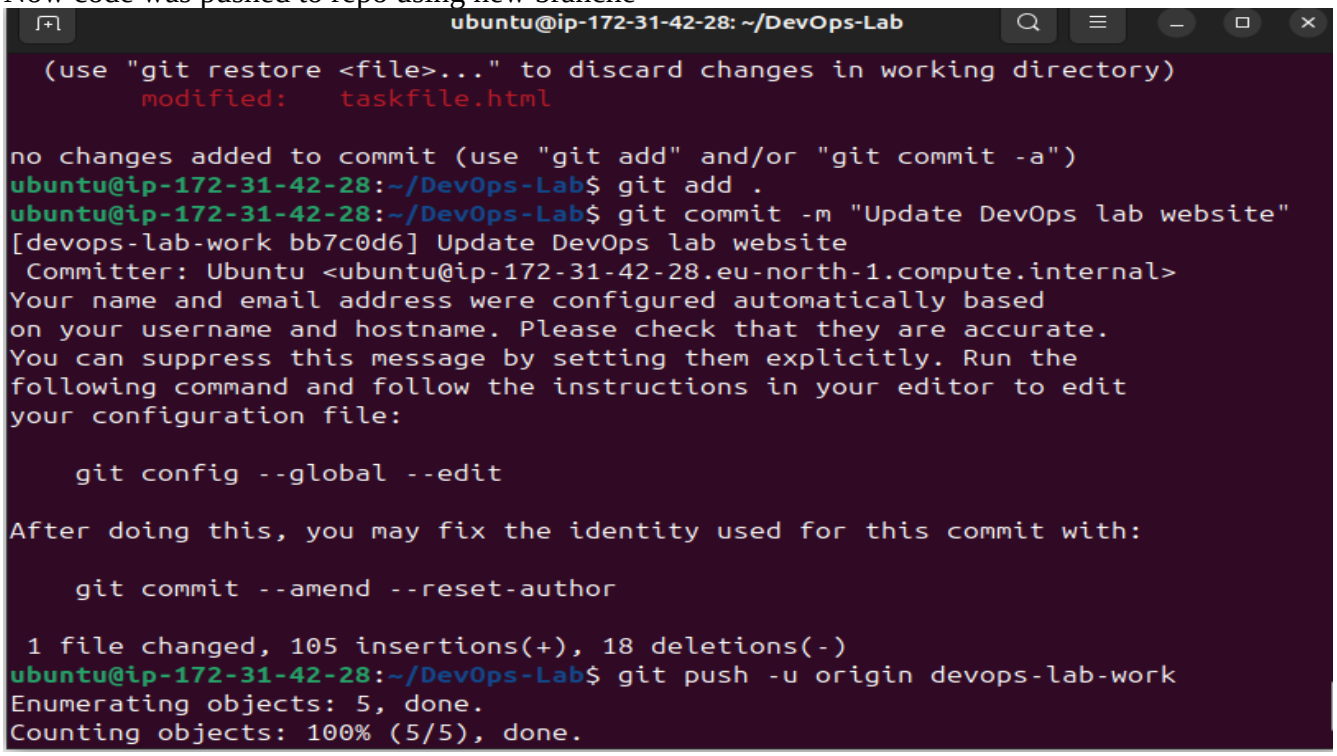
Now we have created new branch from where we will push code and merge it.



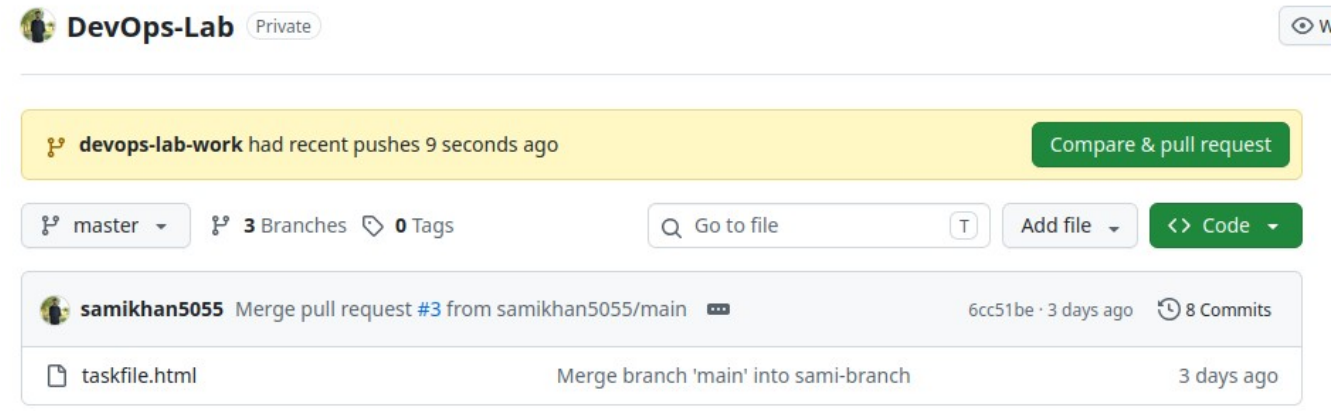
Now we have changed the file using nano and show its status



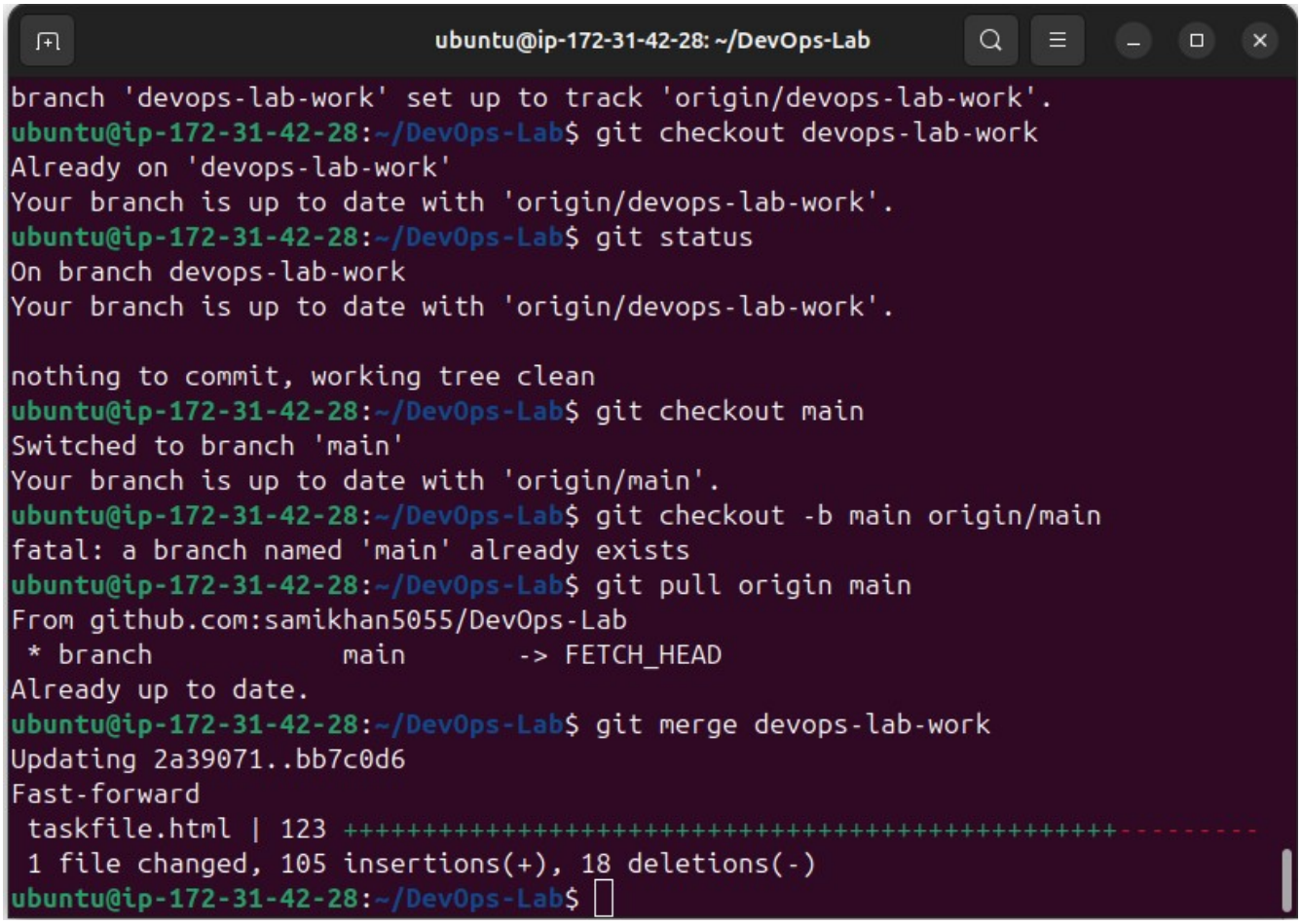
Now code was pushed to repo using new branche



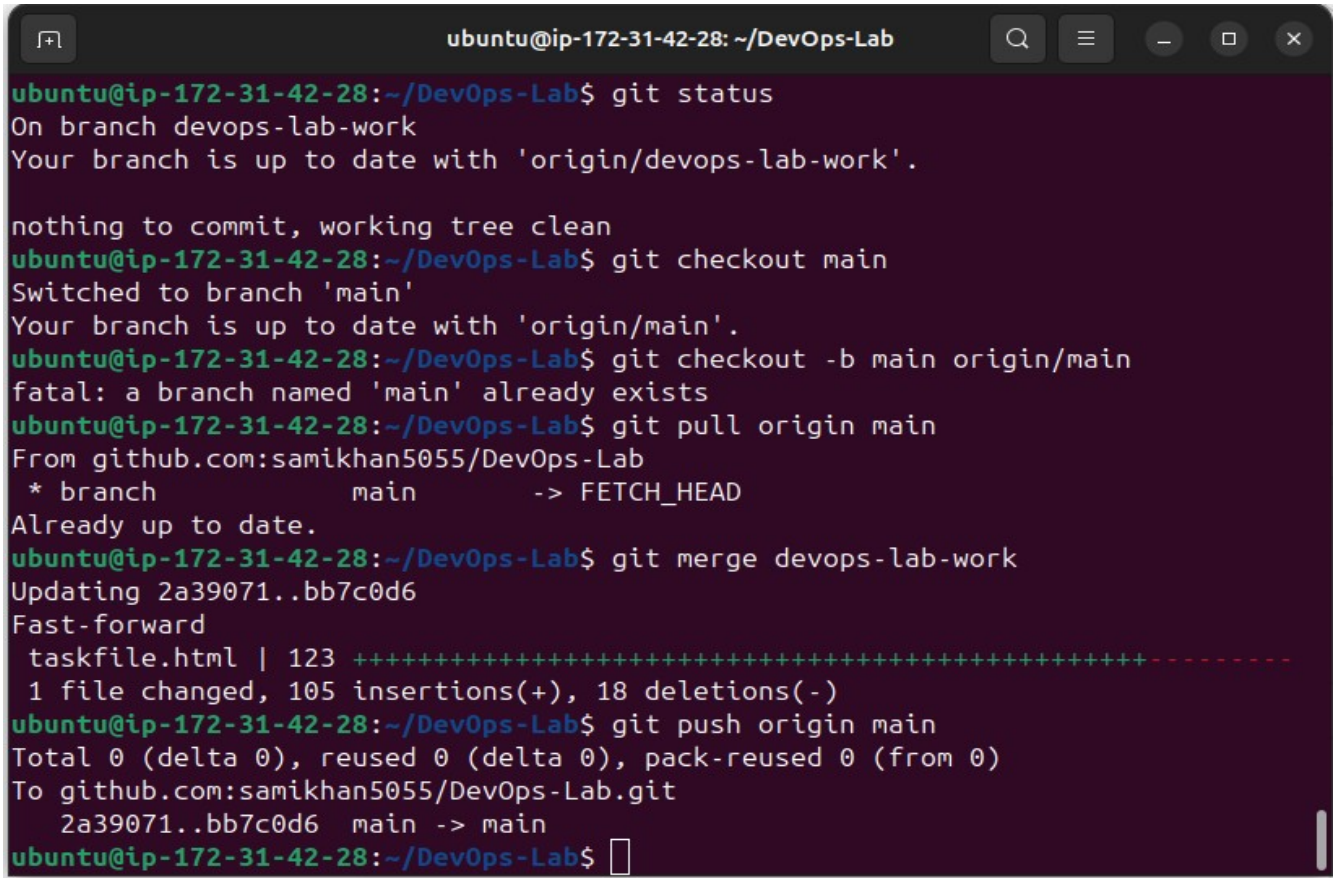
Now it show latest changes on github browser



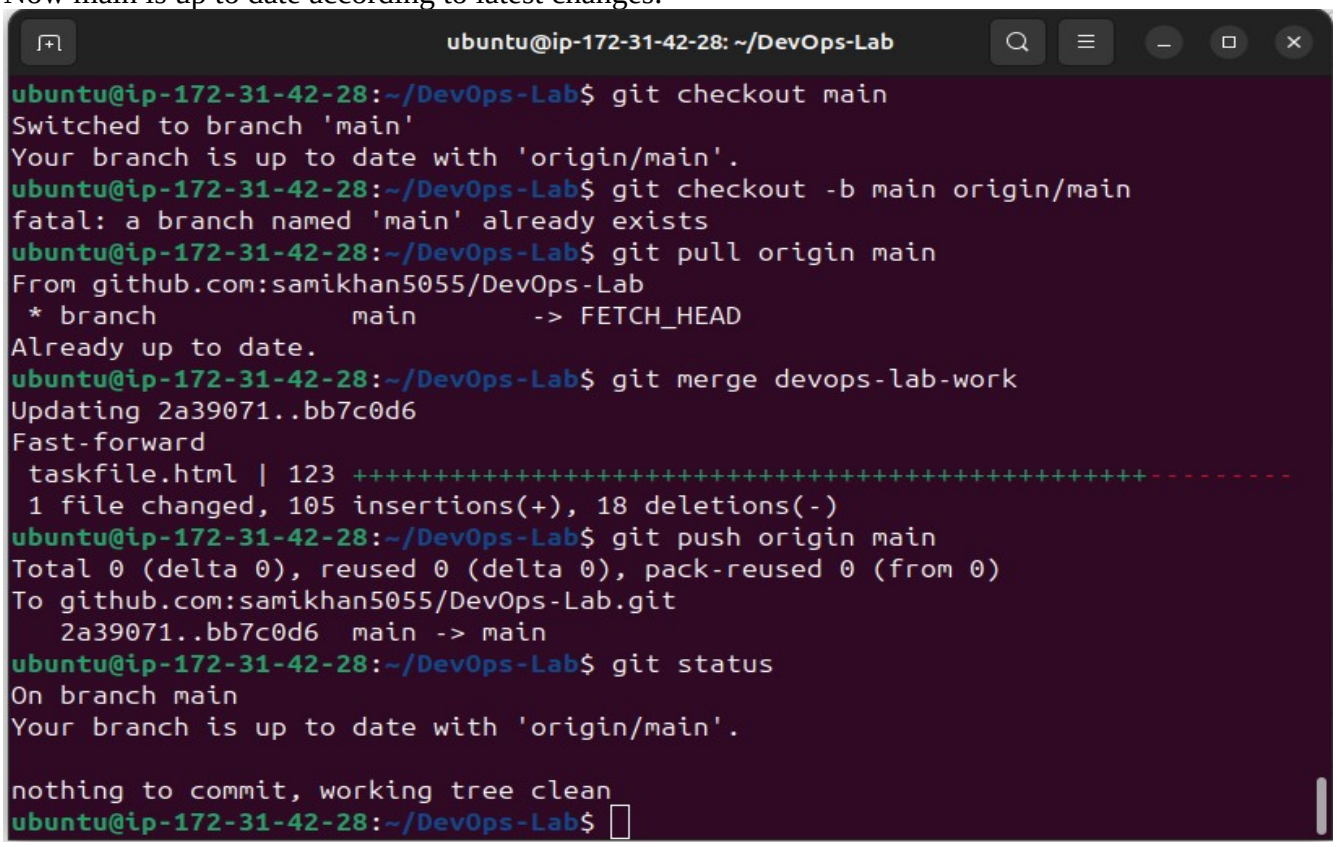
Now we have merged code using cloud vm ubuntu from other branch to main.



Merge changes were being pushed from cloud to github repo using terminal



Now main is up to date according to latest changes.



Now we have all the history of commits and pull requests being done through cloud vm terminal.

Commits

master

All users | All time

Commits on Sep 14, 2026

Merge pull request #4 from samikhan5055/devops-lab-work
samikhan5055 authored now — Verified — f2c96dd

Update DevOps lab website
Ubuntu committed 10 minutes ago — bb7c0d6

Commits on Sep 11, 2026

Merge pull request #3 from samikhan5055/main
samikhan5055 authored 3 days ago — Verified — 6cc51be

Merge pull request #2 from samikhan5055/sami-branch
samikhan5055 authored 3 days ago — Verified — 2a39071

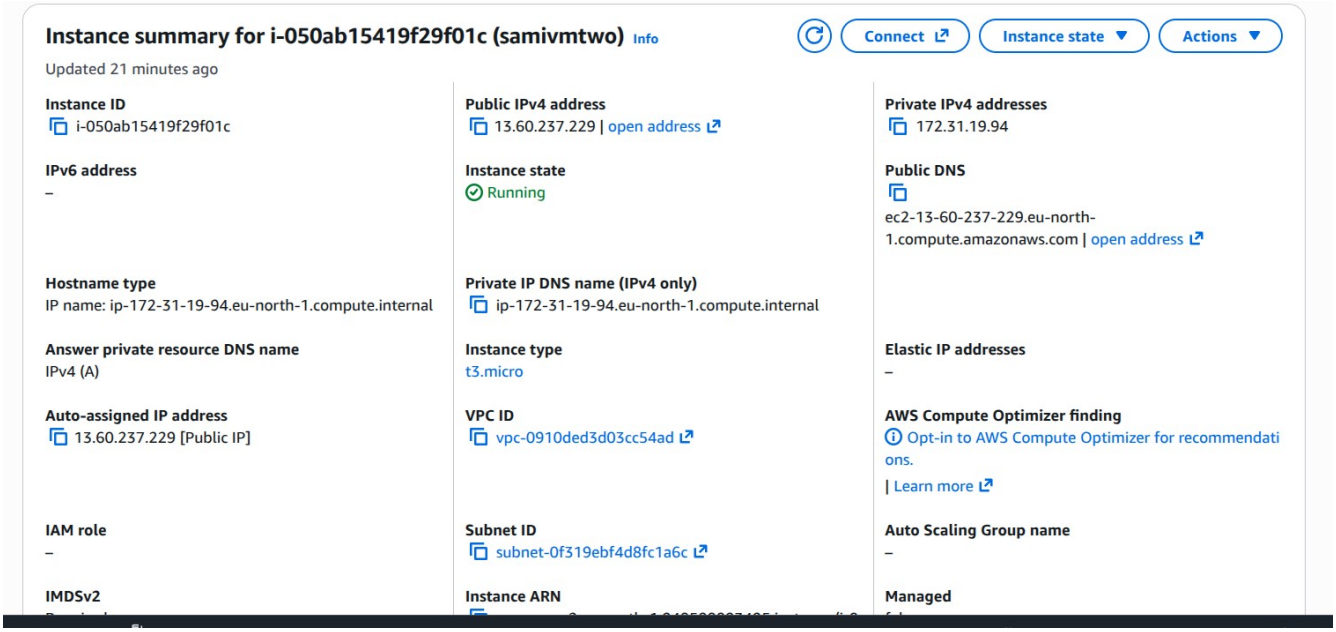
Merge branch 'main' into sami-branch
samikhan5055 authored 3 days ago — Verified — ac43aa6

Website Deployment on AWS EC2 Using Nginx

This report documents the complete deployment of a static hostel accommodation website on an **Amazon EC2** Ubuntu instance using the **Nginx web server.** Every step is shown with a screenshot, and each screenshot is preceded by a short note explaining what it shows and why that step is needed.

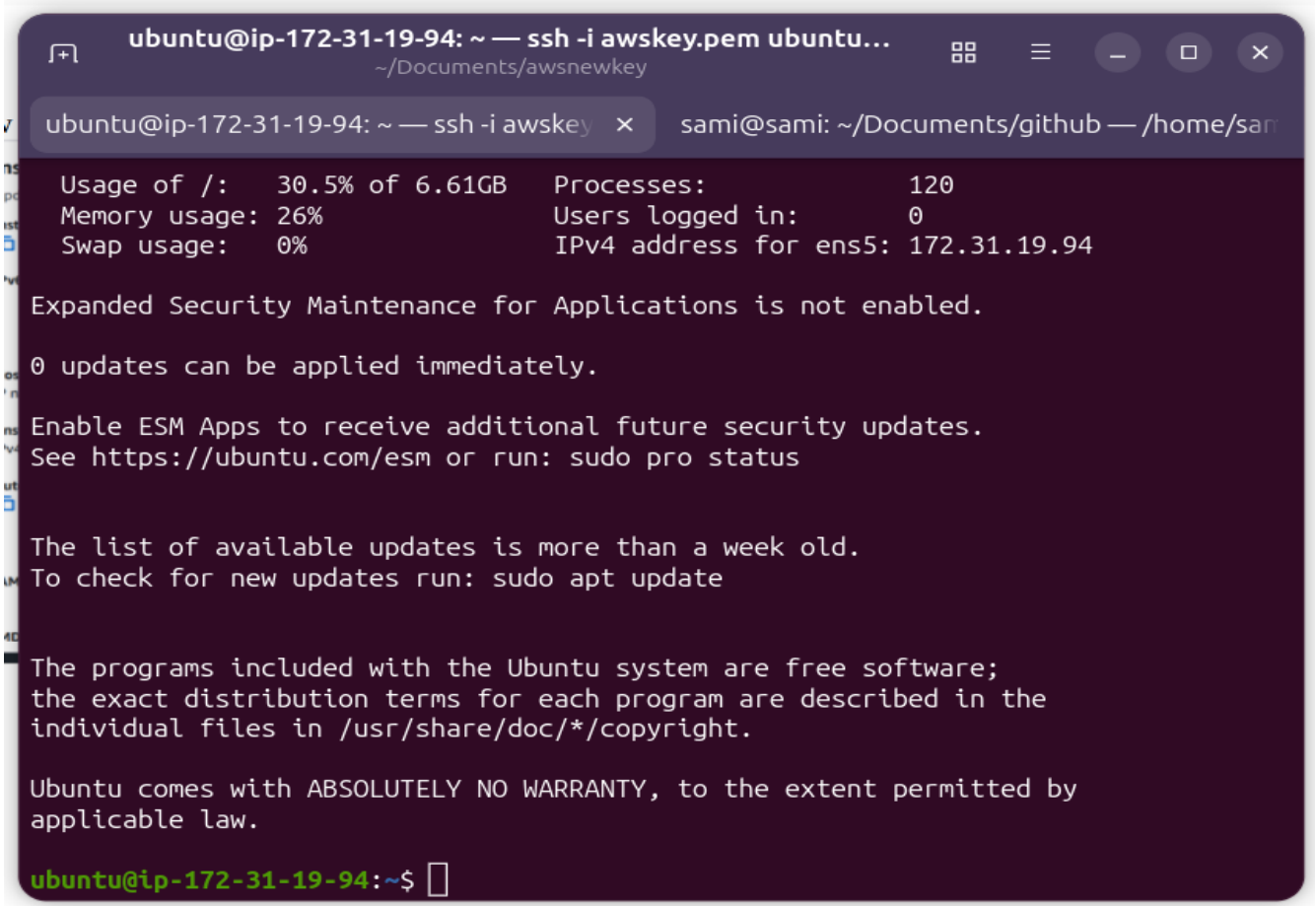
1. Launching the EC2 Instance

A new EC2 instance named `samivmtwo` has been created for completing the website deployment task. The instance summary confirms that it is in the **Running** state and gives the public IPv4 address **13.60.237.229**. This machine acts as the cloud server on which the website will finally be hosted.



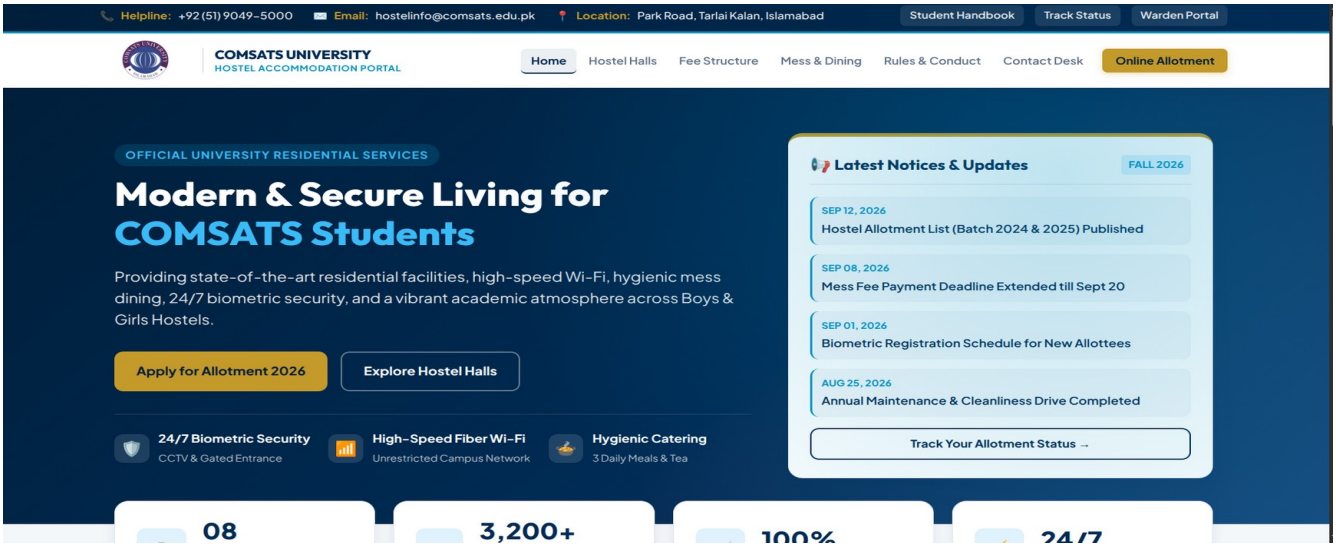
2. Connecting to the Cloud VM

Here we are connected to the cloud VM from the local terminal using SSH with the `awskey.pem` key file. The Ubuntu welcome banner confirms that the login was successful. From this point onwards every command is executed on the EC2 machine and not on the local system.



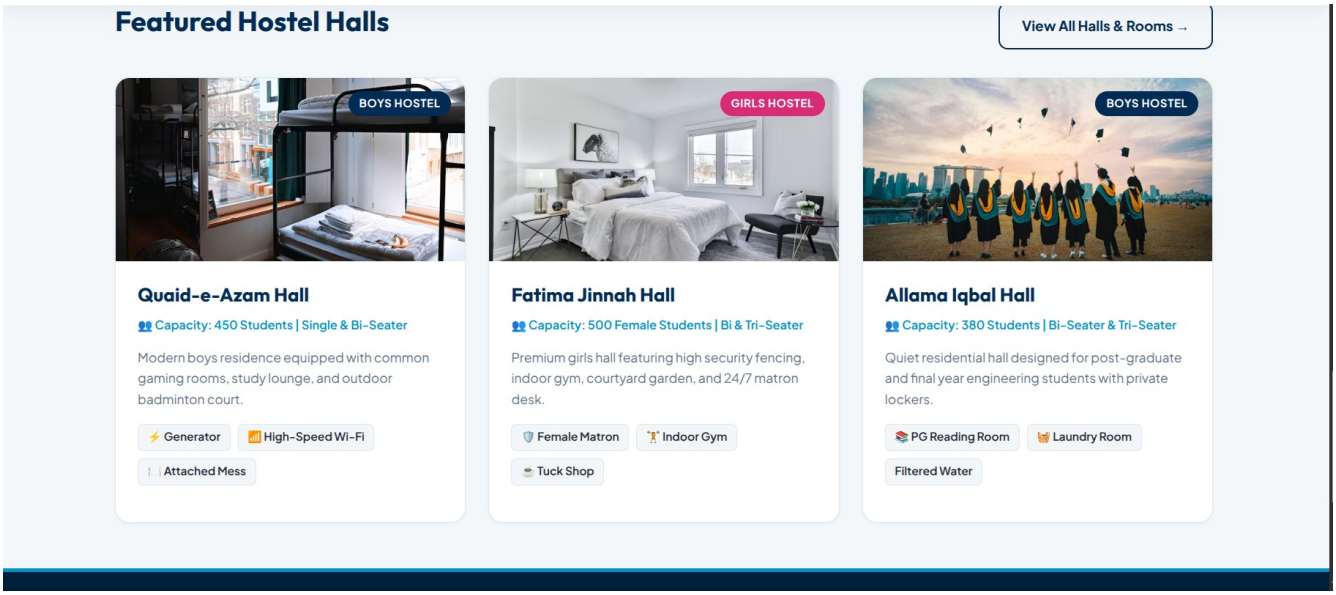
3. Website Frontend Developed and Pushed to GitHub

This is the **frontend** of the website that was developed locally and pushed to a GitHub repository. It is the COMSATS hostel accommodation portal, built with plain HTML and CSS. It is shown here to record what the site looks like before it is deployed on the server.

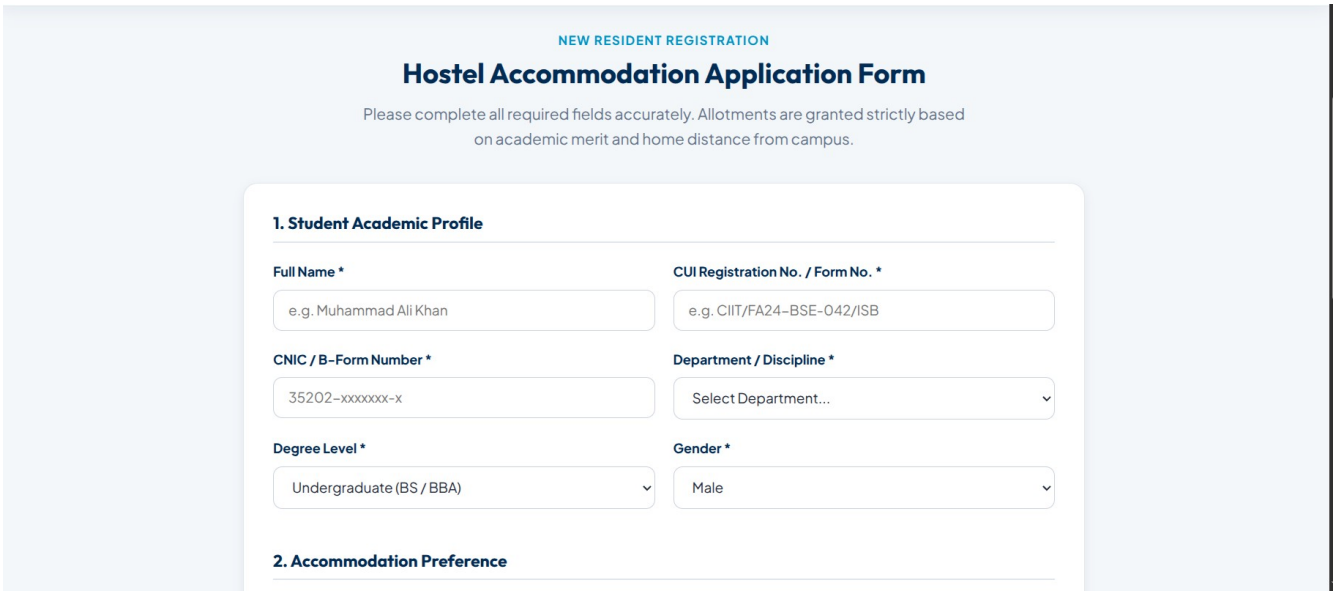


Second page screenshots:

A few more pages of the developed site are shown here. This section lists the featured hostel halls along with their capacity and facilities. It is included to show that the project contains several linked pages and not just a single index file.

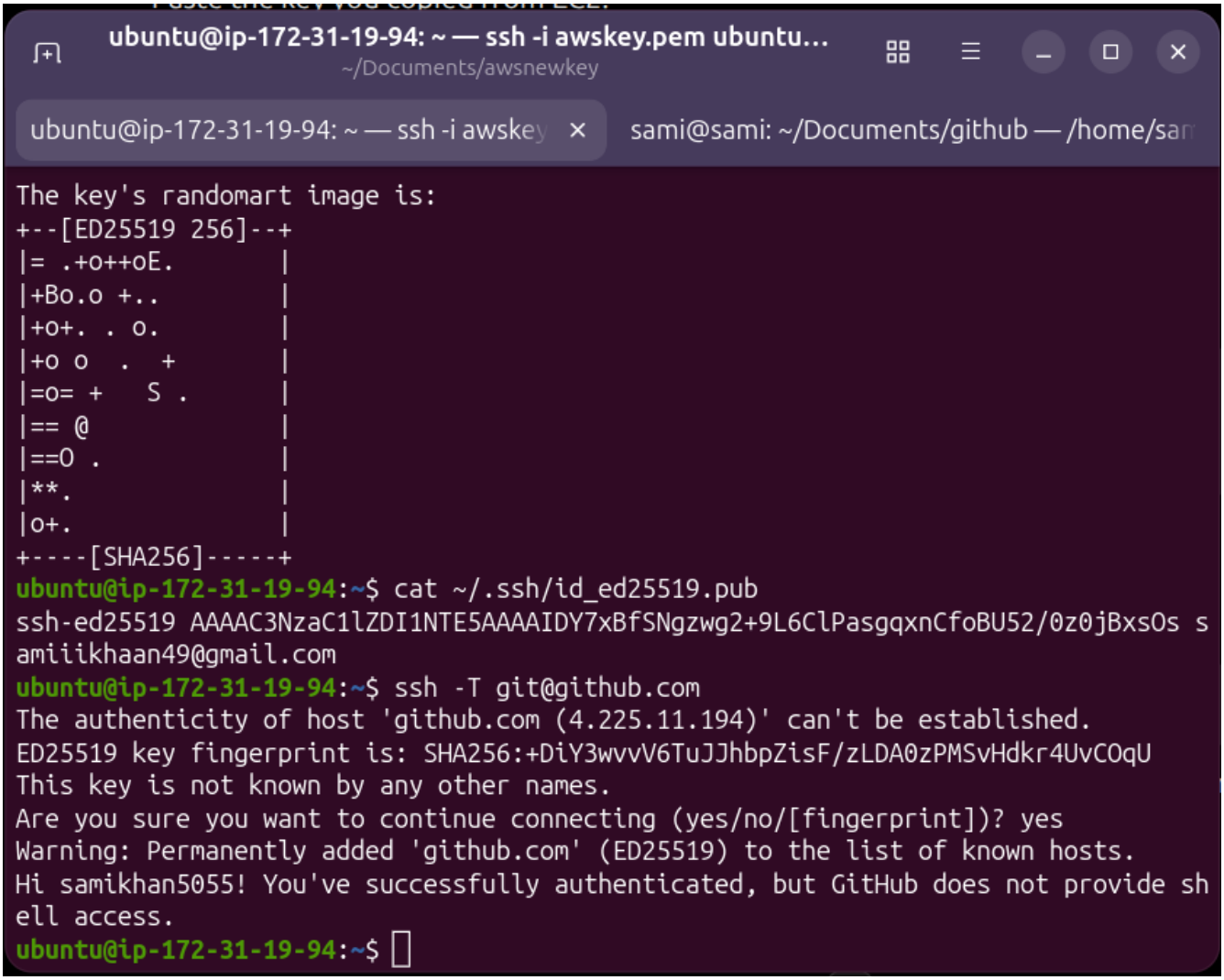


This is the application form frontend page for hostel allotment. It collects the student academic profile and the accommodation preference details. It is shown so that the same page can later be verified once the site is live on the server.

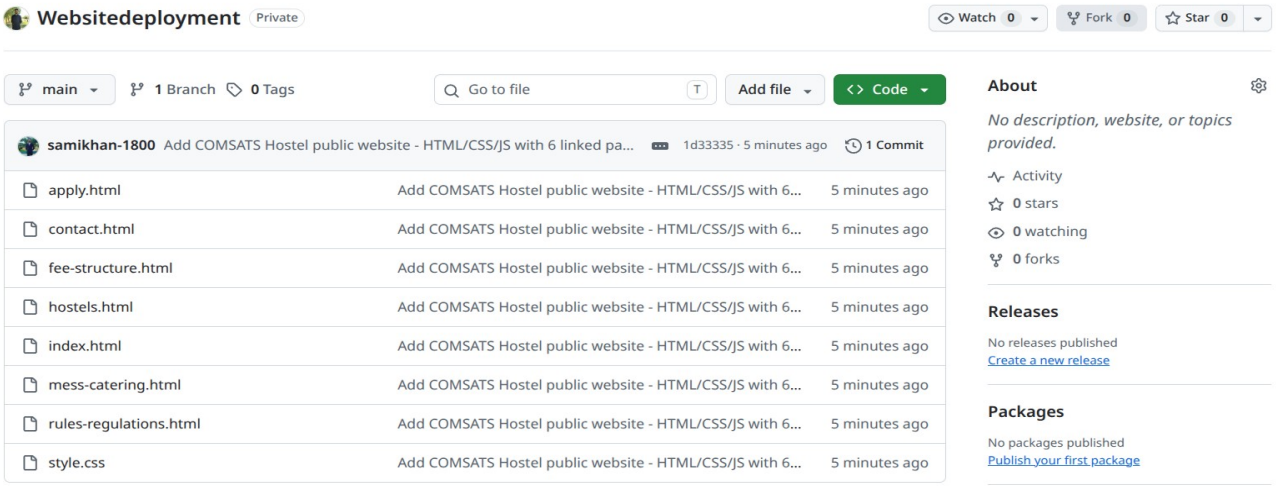


4. Connecting GitHub with the EC2 Instance

Now we have connected GitHub to the EC2 machine from the Ubuntu terminal. The public SSH key is printed with `cat ~/.ssh/id_ed25519.pub` and added to the GitHub account, then **`ssh -T git@github.com`** is run to test it. The success message confirms the server can now reach the repository without entering a password.

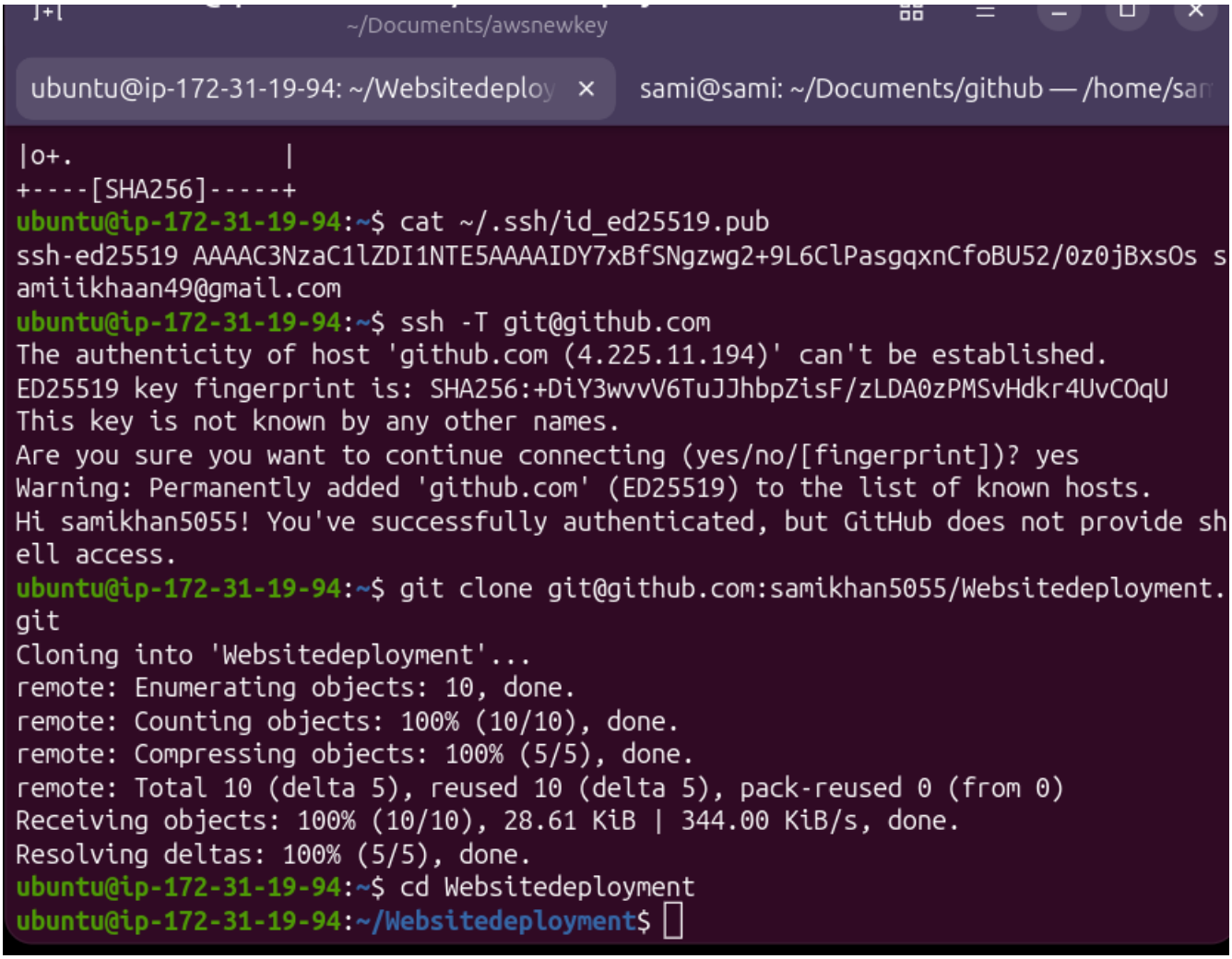


This is the newly created repository `Websitedeployment` containing all the files of the site that will be deployed. Pages such as `index.html`, `apply.html`, `contact.html` and `style.css` are visible in the listing. This is the repository that will be cloned onto the cloud VM.

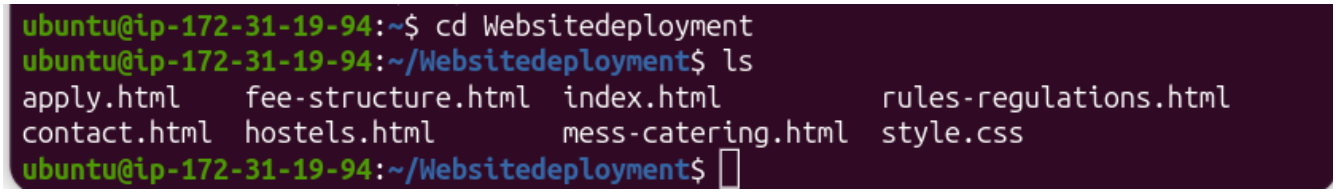


5. Cloning the Repository on the Server

The repository is being cloned on the cloud VM using **`git clone` over SSH**, and then we change into the `Websitedeployment` directory. Cloning is used so that exactly the same code that is on GitHub is placed on the server. This avoids uploading files manually and keeps the two copies in sync.



Now we are displaying all the files inside the repository that was cloned, using the **`ls`** command. The listing shows every HTML page along with the stylesheet. This check is done to make sure the whole project was downloaded correctly before copying it into the web server directory.



6. Installing and Configuring the Nginx Web Server

Following commands were run in the terminal for the setup of the web server.

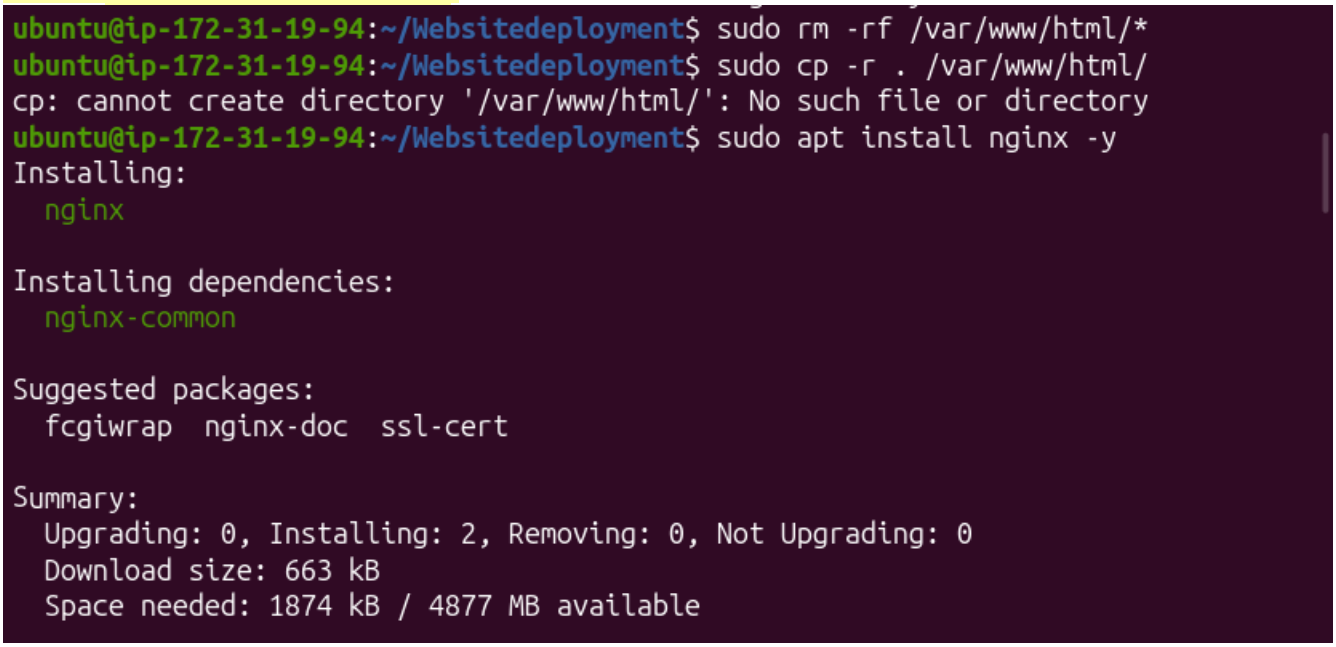
`sudo rm -rf /var/www/html/*`

This command removes the default files from Nginx's web directory. It clears the existing Nginx welcome page so that it can be replaced with our own website.

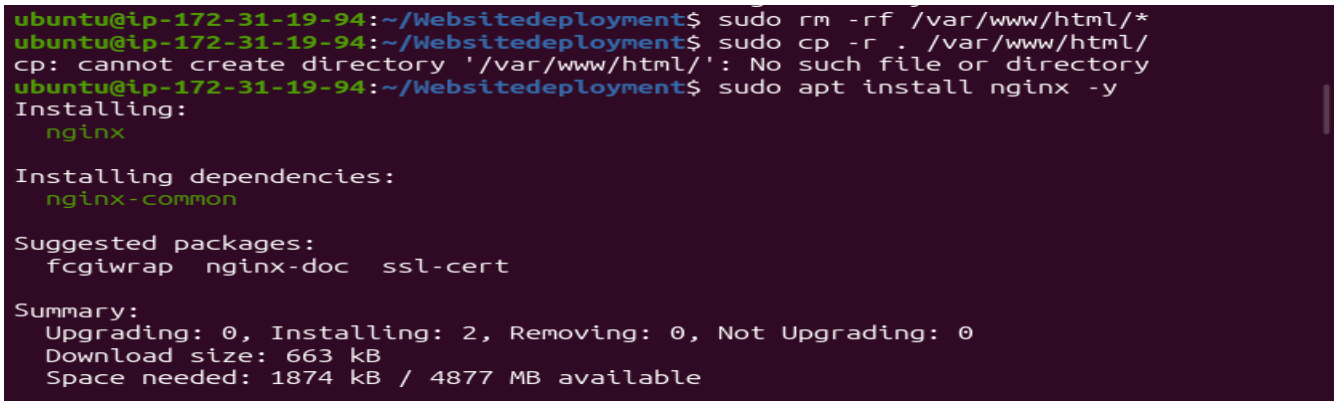
`sudo cp -r . /var/www/html/`

This command copies all files from the current `Websitedeployment` GitHub folder into Nginx's web directory. The HTML, CSS, JavaScript and other website files are therefore placed where Nginx can serve them.

This screenshot shows `sudo apt install nginx -y` being run, which installs the **Nginx web server** on the EC2 Ubuntu machine. Nginx is required because the EC2 instance has no web server of its own and cannot serve pages without one. The installation completed successfully with Nginx version `1.28.3-2ubuntu1.11`.



This installs the **Nginx web server** on the EC2 Ubuntu machine. The installation completed successfully with Nginx version `1.28.3-2ubuntu1.11`.



Here `sudo systemctl status nginx` is used to check the state of the web server service. The output shows the service as **active (running)** and enabled at boot. This confirms that Nginx is live and ready to serve files from `/var/www/html`.

```
No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-19-94:~/Websitedeployment$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enab>
     Active: active (running) since Tue 2026-09-15 08:49:10 UTC; 25s ago
 Invocation: 4c951a02e2104de9922dae44b16cb3cb
       Docs: man:nginx(8)
    Process: 2024 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process>
    Process: 2026 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (cod>
   Main PID: 2055 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3.1M (peak: 7.3M)
        CPU: 43ms
     CGroup: /system.slice/nginx.service
             ├─2055 "nginx: master process /usr/sbin/nginx -g daemon on; master_p>
             ├─2058 "nginx: worker process"
             └─2059 "nginx: worker process"

Sep 15 08:49:10 ip-172-31-19-94 systemd[1]: Starting nginx.service - A high perfo>
Sep 15 08:49:10 ip-172-31-19-94 systemd[1]: Started nginx.service - A high perfor>
lines 1-18/18 (END)
```

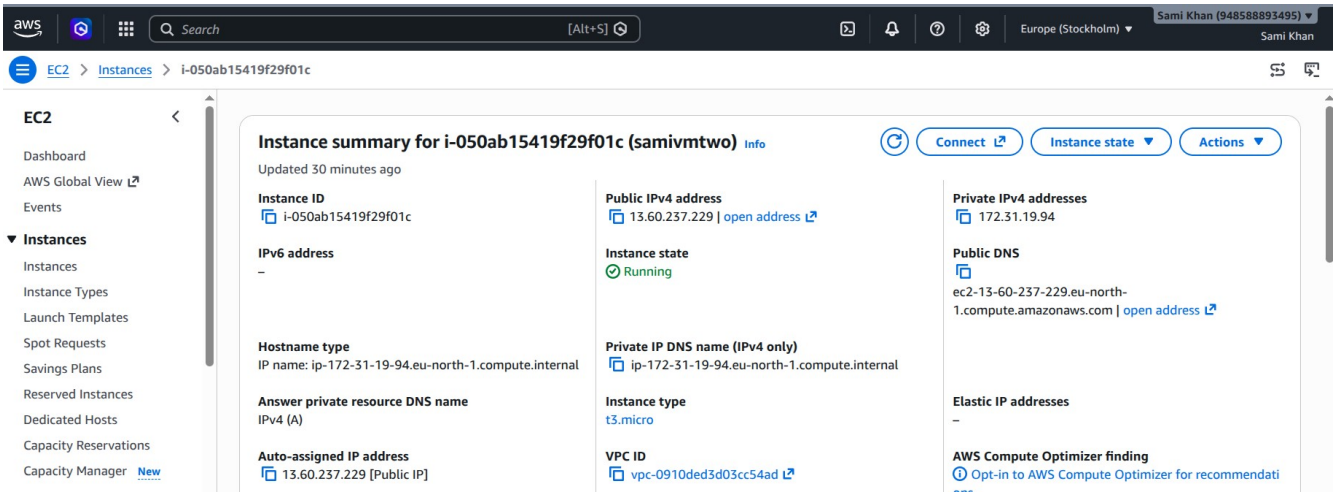
Initially files were not copied when I was opening public ip in browser it was showing nginx default page then default files were removed and actual repo files were copied to nginx.

```
ubuntu@ip-172-31-19-94: ~/Websitedeployment — ssh -...
~/Documents/awsnewkey
ubuntu@ip-172-31-19-94: ~/Websitedeploy ×   sami@sami: ~/Documents/github — /home/sar

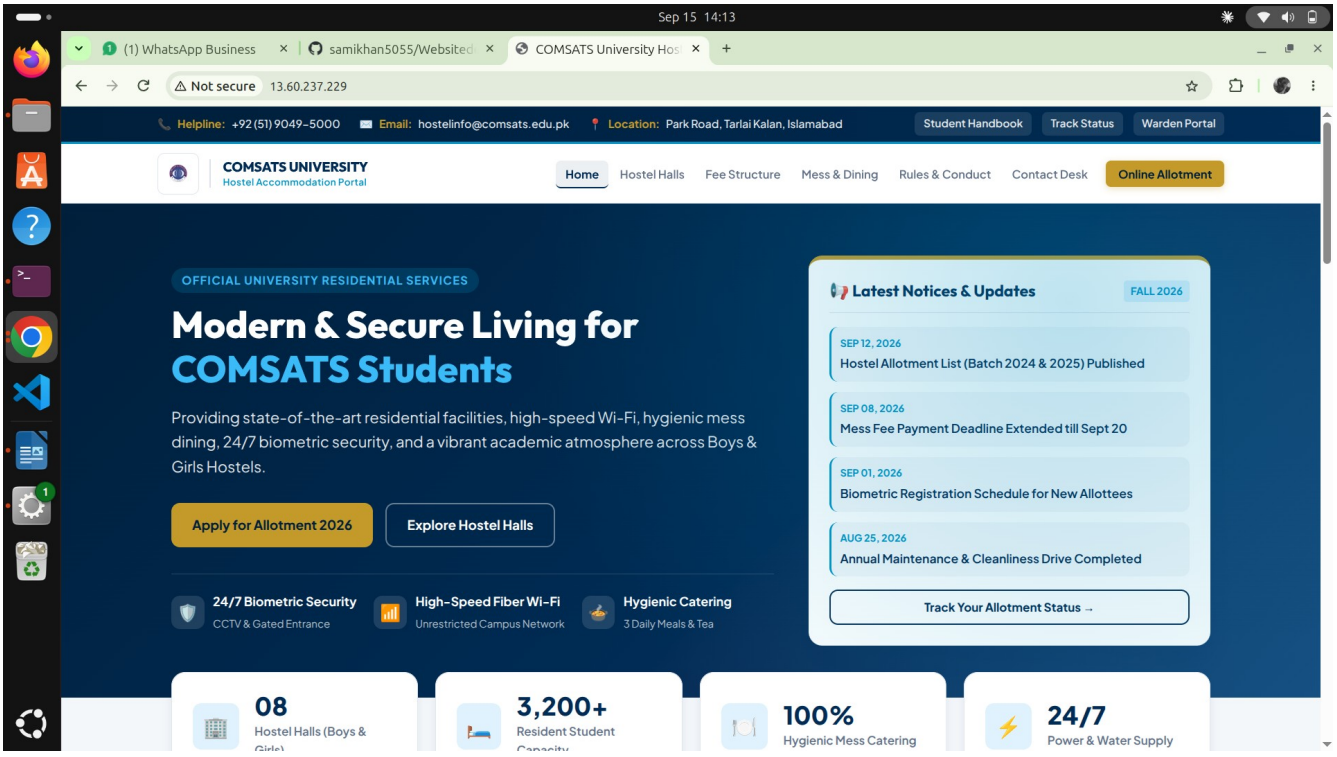
Sep 15 08:57:49 ip-172-31-19-94 systemd[1]: Starting nginx.service - A high perfo>
Sep 15 08:57:49 ip-172-31-19-94 systemd[1]: Started nginx.service - A high perfor>
ubuntu@ip-172-31-19-94:~/Websitedeployment$ ls -la /var/www/html/
total 12
drwxr-xr-x 2 root root 4096 Sep 15 08:49 .
drwxr-xr-x 3 root root 4096 Sep 15 08:49 ..
-rw-r--r-- 1 root root  615 Sep 15 08:49 index.nginx-debian.html
ubuntu@ip-172-31-19-94:~/Websitedeployment$ sudo rm -rf /var/www/html/*
ubuntu@ip-172-31-19-94:~/Websitedeployment$ ls
apply.html    fee-structure.html  index.html          rules-regulations.html
contact.html  hostels.html        mess-catering.html  style.css
ubuntu@ip-172-31-19-94:~/Websitedeployment$ sudo cp -r . /var/www/html/
ubuntu@ip-172-31-19-94:~/Websitedeployment$ ls -la /var/www/html/
total 160
drwxr-xr-x 3 root root  4096 Sep 15 09:01 .
drwxr-xr-x 3 root root  4096 Sep 15 08:49 ..
drwxr-xr-x 7 root root  4096 Sep 15 09:01 .git
-rw-r--r-- 1 root root 14901 Sep 15 09:01 apply.html
-rw-r--r-- 1 root root 13743 Sep 15 09:01 contact.html
-rw-r--r-- 1 root root 15633 Sep 15 09:01 fee-structure.html
-rw-r--r-- 1 root root 21688 Sep 15 09:01 hostels.html
-rw-r--r-- 1 root root 20220 Sep 15 09:01 index.html
-rw-r--r-- 1 root root 18117 Sep 15 09:01 mess-catering.html
-rw-r--r-- 1 root root 11593 Sep 15 09:01 rules-regulations.html
```

7. Verifying the Deployed Website

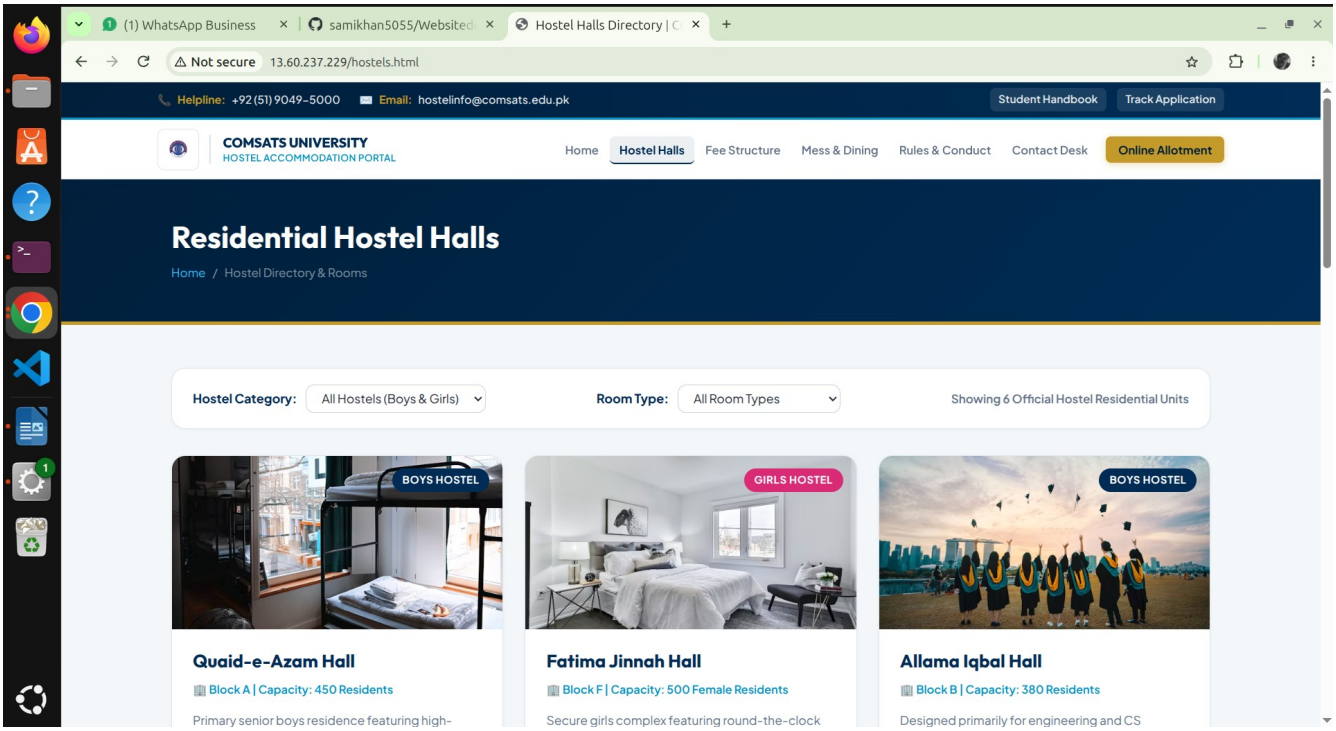
The EC2 instance is running and the console is showing the public IP from which we will open our site in the browser, which is **13.60.237.229**. This is the only address through which the deployed site can be reached. It is checked here because a stopped instance would make the site unreachable.



Now we have opened our deployed site in the browser using the link `http://13.60.237.229/`. The homepage loads correctly, which means Nginx is serving the files from `/var/www/html`. This confirms that the site is successfully deployed on Nginx and running in the browser.

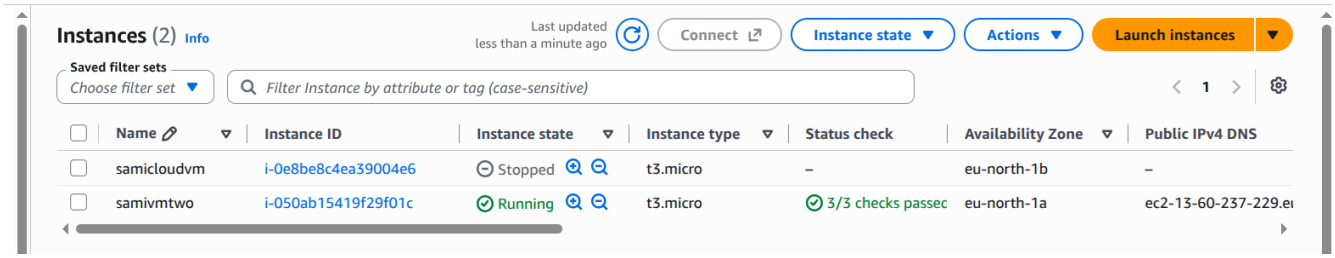


The hostel halls page loads with all of its styling and images intact. This is shown to confirm that the internal links and the CSS file are also being served properly and not only the homepage.



8. Final State of the EC2 Instances

And in the end this screenshot shows the current instance that is running and the other one that is closed. The running instance is the one serving the deployed website on `13.60.237.229`. It is included to record the final state of the resources at the end of the task.



Click on provided text to open the site; **COMSATS HOSTEL**